

RAL-MoE-ROM: HIERARCHICAL MIXTURE-OF-EXPERTS FOR REGIME-ADAPTIVE REDUCED-ORDER MODEL

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ABSTRACT

Reduced-order models (ROMs) approximate high-dimensional dynamical systems using a small number of state variables. However, their predictive accuracy can deteriorate when parameter changes lead to different dynamical regimes. Different regimes may require different reduced representations and dynamics, making it difficult for a single model to remain accurate across the full parameter range. We introduce RAL-MoE-ROM, a hierarchical mixture-of-experts framework with two levels: regime-local specialists and cross-regime routing with fusion.. At the first level, each local model, called a specialist, uses a structured sparse mixture of experts to improve its reduced dynamics and is trained through multi-step autonomous rollouts. At the second level, an outer router selects a specialist or a pair of neighboring specialists. When two specialists are selected, a learned gate uses the initial physical history to combine their reconstructed fields. Across three flow benchmarks spanning steady, Hopf-onset, and periodic regimes, the local specialists provide accurate autonomous predictions, while cross-regime routing further improve prediction in the evaluated regime overlaps.

1 INTRODUCTION

High-fidelity simulations of high-dimensional dynamical systems can be expensive when repeated predictions are required across many parameter settings. Reduced-order models (ROMs) provide a cost-effective surrogate by representing the physical state with a small number of state variables. The problem becomes more challenging when changing a parameter also changes the type of regimes. In incompressible flows, for example, varying the Reynolds number can lead from relaxation toward a steady state to oscillatory growth and sustained periodic motion. A useful predictive model must capture these regimes while evolving autonomously.

Such regime variation introduces two closely related challenges for ROMs. The first concerns representation: spatial structures that are important in one regime may differ from those required in another, making a single representation less effective. The second is dynamical: the reduced evolution law must accommodate qualitatively different behaviors, including stable relaxation, instability growth, and nonlinear periodic oscillations. Increasing model capacity alone cannot compensate for an inadequate reduced representation, while improving the representation does not by itself guarantee accurate autonomous rollout. These observations motivate adapting the local representation and the local dynamics separately, while providing a consistent way to assemble their predictions.

Projection-based ROMs provide an explicit reduced representation and reduced dynamics, commonly through proper orthogonal decomposition (POD) and Galerkin projection (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). However, projection-based ROMs are inherently affected by truncation, which neglects the influence of discarded modes on the retained coordinates, leading to closure errors that degrade long-horizon prediction. (Amsallem & Farhat, 2012). Data-driven ROMs offer greater flexibility in approximating nonlinear dynamics, but low one-step error does not guarantee accurate autonomous rollouts, and they often lack physical interpretability (Duraissamy et al., 2019; Brunton et al., 2020). Hybrid ROMs combine projection-based ROMs with learned corrections or learned reduced representations, retaining physical structure while increasing approximation flexibility (Dar et al., 2023; Lee & Carlberg, 2020; Fresca & Manzoni, 2022; Geelen et al.,

2023). These developments improve the modeling of reduced dynamics, but they do not by themselves resolve the representational difficulty that arises when qualitatively different regimes must be described over a broad parameter range.

A complementary strategy localizes the reduced representation, using multiple subspaces, clusters, or manifolds for different portions of the solution set (Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These directions address complementary aspects of the broad-parameter problem: hybrid modeling improves the fidelity of the reduced dynamics, whereas localization adapts the reduced representation to regime-dependent structures. Localization, however, creates a new assembly problem. Independently constructed local ROMs generally have different reduced state variables and operators, so their reduced coordinates are chart-specific. Direct interpolation or averaging in reduced-coordinate space therefore has no natural cross-chart interpretation. Mixture-of-experts (MoE) architectures provide a natural framework for conditional specialization (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017; Fedus et al., 2022), but standard MoE formulations typically route specialized predictors within a shared representation or common feature space. Here, in contrast, the experts are complete local ROMs whose coordinates are not shared. The problem is therefore not only routing among local ROMs, but also assembling or fusing their outputs across representations. This makes the problem hierarchical: one must adapt the reduced dynamics within each local representation and, separately, route to and fuse complete local ROMs across representations. This raises a distinct question that standard within-representation MoE routing does not address: how can independently constructed local dynamical models be routed and assembled without forcing them into a common latent coordinate system?

To address this gap, we introduce the Regime-Adaptive Localized Mixture-of-Experts Reduced-Order Model (RAL-MoE-ROM), a two-level hierarchical framework that separates regime-local reduced dynamics from cross-regime routing and fusion. The parameter domain is covered by prescribed, overlapping regime-local regions, each equipped with an independent autonomous specialist. At the first level, each specialist uses an explicit projected-based ROM and augments it with sparsely structured MOEs which trained through multi-step autonomous rollouts. Rather than using conventional MLP-only experts, each activated expert decomposes its output into three additive components: a nonlinear neural branch, together with explicit linear and low-rank quadratic branches derived from the projection-based. At the second level, a parameter-conditioned outer router produces window-level routing weights over the local specialists, while a fixed applicability mask determines whether inference uses one specialist or a pair of adjacent specialists. When two specialists are selected, a learned fusion gate combines their reconstructed physical fields conditioned on the initial physical history. The overall architecture is illustrated in Figure 1, highlighting the regime-local specialists and the cross-regime routing and fusion mechanism.

Our main contributions are as follows:

- **Structured regime-local specialists for autonomous reduced dynamics.** Each specialist combines a physics-based ROM with a structured sparse MoE closure and is trained under closed-loop multi-step rollouts.
- **Cross-regime routing and fusion.** An outer router selects a specialist or a pair of neighboring specialists. When two specialists are selected, a learned gate uses the initial physical history to combine their reconstructed fields.
- **Multi-regime evaluation.** We evaluate the framework on incompressible-flow benchmarks spanning steady, Hopf-onset, and periodic regimes. The experiments assess autonomous specialist rollouts, cross-regime routing and fusion in regime overlaps, and the main local architectural choices, providing complementary evidence for the proposed local modeling and assembly strategy.

2 RELATED WORK

Projection, learned dynamics, and autonomous rollout. POD–Galerkin methods project the governing equations onto a reduced basis and retain an explicit reduced dynamical system (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). Their long-horizon accuracy can deteriorate when truncated interactions are not adequately represented, motivating closure and stabilization strategies (Amsallem & Farhat, 2012). Non-intrusive and data-driven ROMs instead infer reduced evolution

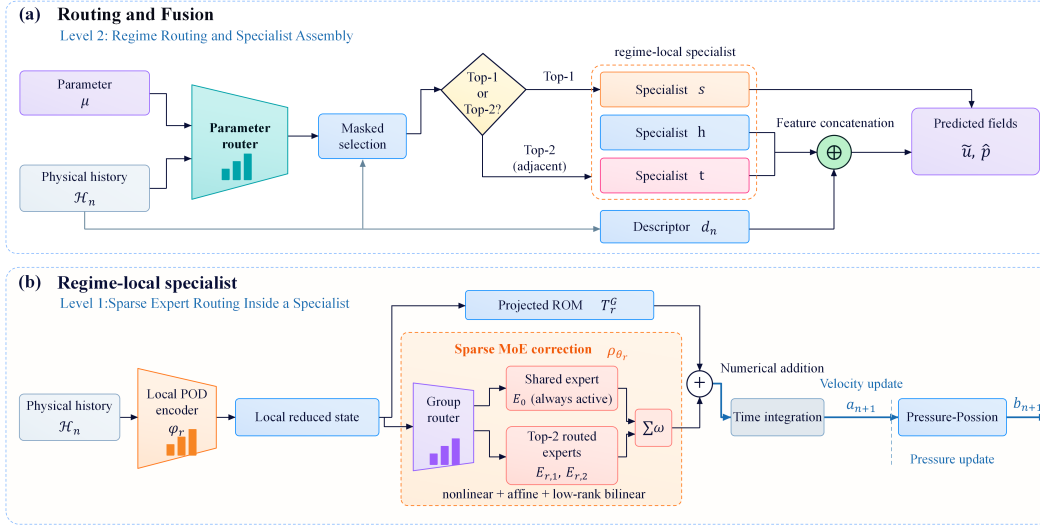


Figure 1: Overview of RAL-MoE-ROM. (a) A parameter-only outer router provides trajectory-level preferences over regime-local specialists, while an applicability mask restricts inference to a single specialist or an adjacent pair. When an adjacent pair is used, a history-conditioned gate combines only their reconstructed physical fields. (b) Each regime-local specialist augments a projected ROMs with a structured sparse MoE correction.

laws directly from simulation data, ranging from projection-based operator inference to neural and latent-state dynamics (Peherstorfer & Willcox, 2016; Hesthaven & Ubbiali, 2018; Champion et al., 2019; Lee & Carlberg, 2020; Pan & Xiao, 2024; Park et al., 2024). Operator-inference methods provide a particularly direct connection between data-driven reduced dynamics and classical projection-based ROMs (Peherstorfer & Willcox, 2016). Neural-operator approaches learn mappings between function spaces rather than explicit low-dimensional state evolution (Lu et al., 2021; Kovachki et al., 2023). Hybrid ROMs retain projected or physics-based structure while learning missing or unresolved reduced dynamics (Duraismy et al., 2019; Fresca & Manzoni, 2022; Dar et al., 2023). RAL-MoE-ROM follows this hybrid principle, but places a structured sparse-MoE correction inside each regime-local ROM and trains the resulting specialist under multi-step autonomous rollout.

Localized representations and local reduced dynamics. Localization has been explored through parameter-dependent reduced bases, localized operators and nonlinear manifolds (Amsallem & Farhat, 2008; Peherstorfer et al., 2014; Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024; Bhat et al., 2025). These approaches all reduce the burden on a single global representation, differing in whether localization is performed along parameter, state, or spatial coordinates. Recent work has further emphasized explicitly local reduced dynamics. More recently, ql-ROM partitions the solution manifold into clusters, constructs an intrusive ROM within each cluster, and connects the local models through assignment and change-of-basis operations (Colanera & Magri, 2025). Therefore, when complete local ROMs are constructed independently, their reduced coordinates and operators need not be directly compatible, so switching or combining them requires an explicit coordination mechanism. RAL-MoE-ROM takes a different approach: instead of coordinating local models in reduced-coordinate space, it lets local specialists evolve independently in their native coordinates and combines neighboring predictions only after reconstruction in the common physical space.

Mixture-of-experts and cross-chart assembly. Classical hierarchical mixtures learn conditional expert assignments (Jacobs et al., 1991; Jordan & Jacobs, 1994), while sparse MoE architectures activate only a subset of experts for each input (Shazeer et al., 2017; Fedus et al., 2022). MoE-based conditional specialization has also been introduced into scientific machine learning, including physics-informed constraints, neural-operator pretraining, and PDE forecasting (Chalapathi et al., 2024; Ben-Shabat et al., 2024; Wang et al., 2025; Li et al., 2026). These methods demonstrate the value of conditional specialization, but do not address routing among complete autonomous ROMs.

RAL-MoE-ROM therefore applies router at two levels: sparse experts model unresolved dynamics within each local ROM, while an outer router acts on the complete regime-local specialists. The outer level selects a single specialist or an applicable neighboring pair, with the latter combined through a history-conditioned gate after physical reconstruction.

3 RAL-MoE-ROM

RAL-MoE-ROM separates reduced modeling into three complementary components: regime-local ROMs, autonomous local specialists, and cross-regime router and fusion. These components are organized by a two-level routing hierarchy. The hierarchy separates internal sparse expert routing within each local specialist from outer parameter-only regime routing (E2). For a prescribed adjacent candidate pair, history-conditioned Top-2 fusion (T2-C) combines independently predicted trajectories after reconstruction in physical space.

3.1 REGIME-LOCAL ROMS

We consider parameterized incompressible flows governed by the Navier–Stokes equations,

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu(\mu) \Delta \mathbf{u} + \mathbf{f}(\mu), \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

with the initial and boundary conditions of each benchmark. Here $\mathbf{u}$ and p denote velocity and pressure, μ is the physical parameter, ν is the viscosity, and $\mathbf{f}$ denotes a prescribed forcing term. In our nondimensional benchmarks, $\mu = Re$ and $\nu(\mu) = Re^{-1}$.

Finite-volume spatial discretization gives the semi-discrete momentum equation and incompressibility constraint,

$$\dot{\mathbf{u}} = \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu) \mathbf{u} - \mathcal{N}_h(\mathbf{u}, \mathbf{u}; \mu) - \nabla_h(\mu) p, \quad \text{div}_h(\mu) \mathbf{u} = 0. \quad (2)$$

Here time remains continuous, while $\mathbf{u} \in \mathbb{R}^{N_u}$ and $p \in \mathbb{R}^{N_p}$ denote the discrete field vectors, using the same symbols as their continuous counterparts. The subscript h denote spatial discretization: $\mathcal{L}_h$ and $\mathcal{N}_h$ represent the viscous and convective terms, ∇_h and div_h the gradient and divergence, and $\mathbf{f}_h$ the discrete forcing.

As μ varies, the flow can exhibit different dynamical regimes. Let $r \in \{S, H, P\}$ index steady, Hopf-onset, and periodic regimes, with corresponding overlapping parameter regions $\mathcal{M}_r$. For each region, we construct separate velocity and pressure bases by area-weighted proper orthogonal decomposition (POD) of high-fidelity snapshots. The velocity basis $\Phi_u^{(r)}$ satisfies $\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I$, where M_u is the quadrature matrix. It defines the local encoding and reconstruction

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \hat{\mathbf{u}}^{(r)} = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (3)$$

where $\bar{\mathbf{u}}^{(r)}$ is the local training mean and $\mathbf{a}^{(r)} \in \mathbb{R}^{d_u^{(r)}}$ denotes the local reduced velocity coordinates for regime r , with $d_u^{(r)}$ the number of retained velocity POD modes. Pressure is first mapped to a common weighted zero-mean gauge and then encoded analogously using $\Phi_p^{(r)}$, yielding coefficients $\mathbf{b}^{(r)}$. Appendix A specifies the pressure gauge, quadrature, and weighted POD construction for this semi-discrete system.

Substituting the local velocity and pressure reconstructions into Eq. (2) and projecting its momentum equation onto $\Phi_u^{(r)}$ in the M_u inner product gives the Galerkin equation:

$$\mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) = \mathbf{c}_r(\mu) + \mathbf{A}_r(\mu) \mathbf{a}^{(r)} + \mathbf{H}_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + \mathbf{C}_r^p(\mu) \mathbf{b}^{(r)}, \quad (4)$$

where the terms collect the constant, linear, quadratic, and pressure contributions. Appendix B defines these projected operators and the reduced Pressure–Poisson equation.

3.2 REGIME-LOCAL HYBRID SPECIALISTS

A hybrid specialist augments the chart-local projected Galerkin equation with a learned velocity correction,

$$\dot{\mathbf{a}}^{(r)} = \mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) + \boldsymbol{\rho}_{\theta_r}^u, \quad (5)$$

and reconstructs pressure algebraically after the velocity update.

Local features and two computation paths. Let $\mathbf{g}_n^{(r)} = \mathcal{F}_r^G(\mathbf{a}_n^{(r)}, \mathbf{b}_n^{(r)}; \mu)$. The chart-specific feature map constructs a standardized vector

$$\boldsymbol{\xi}_n^{(r)} = \text{Feat}_r\left((\mathbf{a}_{n-\ell}^{(r)}, \mathbf{b}_{n-\ell}^{(r)}, \mathbf{g}_{n-\ell}^{(r)})_{\ell=0}^2, \mu\right), \quad \mathbf{z}_n^{(r)} = \text{Enc}_r(\boldsymbol{\xi}_n^{(r)}). \quad (6)$$

Features include current states, projected Galerkin equation, and parameter features. Standardization uses only the local training set.

The encoded features feed two paths: expert responses use $[\mathbf{z}; \mathbf{s}^c]$, whereas routing uses $\boldsymbol{\chi} = [\mathbf{z}; \boldsymbol{\xi}]$. $c \in \{u, p\}$ denotes the velocity or pressure output channel, and

$$\mathbf{s}^u = \tilde{\mathbf{a}}^{(r)}, \quad \mathbf{s}^p = [\tilde{\mathbf{a}}^{(r)}; \tilde{\mathbf{b}}^{(r)}] \quad (7)$$

are the current-state slices of the standardized feature vector. Time and chart indices are suppressed below for readability.

Expert-response path. For expert e in group m , a nonlinear residual feed-forward network $N_{e,m}^c$ first projects $[\mathbf{z}; \mathbf{s}^c]$ to its hidden representation, applies expanded residual blocks with GELU activations, and maps the result to d_c outputs, with $d_c = d_u^{(r)}$ or $d_p^{(r)}$. This response is augmented by explicit linear and low-rank quadratic functions of the current state:

$$[E_{e,m}^c(\mathbf{z}, \mathbf{s}^c)]_j = [N_{e,m}^c([\mathbf{z}; \mathbf{s}^c])]_j + [L_{e,m}^c \mathbf{s}^c]_j + \kappa \sum_{\ell=1}^{q_b} (\mathbf{v}_{e,m,j\ell}^{c,(1)})^\top \mathbf{s}^c (\mathbf{v}_{e,m,j\ell}^{c,(2)})^\top \mathbf{s}^c. \quad (8)$$

The linear map has no bias, and the two quadratic factors are independent for each output component j . The verified configurations use $q_b = 4$ and $\kappa = 0.05$. Shared and routed experts have the same architecture but separate parameters. Thus the nonlinear branch uses history-conditioned features, while the explicit polynomial branches act directly on local state coordinates.

Routing-and-weighting path. The group router receives $\boldsymbol{\chi} = [\mathbf{z}; \boldsymbol{\xi}]$ and computes

$$\ell^{\text{grp}} = R_r^{\text{grp}}(\boldsymbol{\chi}), \quad \mathbf{p}^{\text{grp}} = \text{softmax}(\ell^{\text{grp}}/T_g), \quad m^* = \arg \max_m p_m^{\text{grp}}, \quad (9)$$

where $T_g > 0$ is the group temperature. The router is a layer-normalized MLP with a SiLU hidden activation and dropout. Selection is hard Top-1 for each sample. During training, a straight-through gate $\tilde{\mathbf{q}} = \text{onehot}(m^*) - \text{stopgrad}(\mathbf{p}^{\text{grp}}) + \mathbf{p}^{\text{grp}}$ retains hard selection in the forward pass while providing a softmax gradient path. Within the selected group, independent velocity and pressure routers with the same MLP form score six routed experts:

$$\ell^c = R_{r,m^*}^c(\boldsymbol{\chi}), \quad \mathbf{p}^c = \text{softmax}(\ell^c/T_c), \quad \mathcal{T}_2^c = \text{Top2}(\mathbf{p}^c). \quad (10)$$

Here Top2 returns the indices of the two largest scores and $T_c > 0$ is the channel temperature. The shared expert $e = 0$ is excluded from this competition and remains active in the selected group. Given configured shared and routed scales $\lambda_0, \lambda_{\text{rt}} > 0$, the weights are

$$\omega_0^c = \frac{\lambda_0}{\lambda_0 + \lambda_{\text{rt}}}, \quad \omega_e^c = \frac{\lambda_{\text{rt}}}{\lambda_0 + \lambda_{\text{rt}}} \frac{p_e^c}{\sum_{j \in \mathcal{T}_2^c} p_j^c}, \quad e \in \mathcal{T}_2^c. \quad (11)$$

The implementation includes numerical safeguards in the denominators; the expression above gives the normalized mathematical rule. The two paths meet in the sparse mixture

$$\mathcal{M}_{\theta_r}^c = \omega_0^c E_{0,m^*}^c(\mathbf{z}, \mathbf{s}^c) + \sum_{e \in \mathcal{T}_2^c} \omega_e^c E_{e,m^*}^c(\mathbf{z}, \mathbf{s}^c). \quad (12)$$

Expert outputs are combined in standardized target coordinates and then converted using the chart-specific output scalers to the velocity and pressure corrections $\boldsymbol{\rho}_{\theta_r}^u$ and $\boldsymbol{\rho}_{\theta_r}^p$ in the additive instances.

Algebraic pressure reconstruction. In adaptive-gate instances, pressure is reconstructed after the velocity update as

$$\mathbf{b}_{n+1}^{(r)} = \boldsymbol{\gamma}_n^{(r)} \odot \mathcal{Q}_r^{PP}(\mathbf{a}_{n+1}^{(r)}; \mu) + \boldsymbol{\rho}_n^{p,(r)}, \quad \boldsymbol{\gamma}_n^{(r)} = \text{sigmoid}(\boldsymbol{\eta}_n^p(\mathbf{z}_n^{(r)})). \quad (13)$$

The independent confidence head η_r^p maps the shared embedding through a 64-unit GELU hidden layer to $d_p^{(r)}$ outputs. It scales only the PPE candidate defined in Appendix B; the learned correction is additive, not multiplied by $1 - \gamma_n^{(r)}$. Both terms are combined in the same unstandardized local pressure coordinates. The correction $\rho_n^{p,(r)}$ uses the current encoded context; where configured, its expert state input is overridden using the updated velocity and PPE pressure, without re-encoding the routing context. Pressure is therefore reconstructed rather than independently time-integrated; its output enters the subsequent local history.

3.3 CLOSED-LOOP ROLLOUT TRAINING

Each specialist is initialized by encoding a physical history into its own local coordinates. It then advances autonomously, using its predicted velocity and reconstructed pressure to update the history. No reference state inside the prediction window is injected after initialization.

Frozen-context advancement. During one reduced velocity step, the pressure $b_n^{(r)}$, preceding-state history, and parameter define a fixed context $\mathcal{C}_n^{(r)}$. For a trial velocity state $\mathbf{a}$, the hybrid right-hand side and numerical update are

$$F_r(\mathbf{a}; \mathcal{C}_n^{(r)}) = \mathcal{F}_r^G(\mathbf{a}, \mathbf{b}_n^{(r)}; \mu) + \rho_{\theta_r}^u(\xi^{(r)}(\mathbf{a}; \mathcal{C}_n^{(r)})), \quad (14)$$

$$\mathbf{a}_{n+1}^{(r)} = \text{Step}_r(\mathbf{a}_n^{(r)}; F_r(\cdot; \mathcal{C}_n^{(r)}), \Delta t_r). \quad (15)$$

Here Δt_r is the reduced time step and Step_r denotes the chart-specific advancement rule. For continuous-time instances, Runge–Kutta stages evaluate the velocity-dependent right-hand side with the same frozen context; each trial velocity triggers recomputation of the Galerkin right-hand side, current features, embedding, internal group selection, channel Top-2 weights, and expert responses. Intermediate pressure outputs that may be computed by the full network forward pass are not used to advance pressure at the Runge–Kutta stages. The updated velocity is then supplied to Eq. equation 13, and the local history is shifted before the next step. Benchmark-specific stepping rules are given in Appendix D.

Rollout prediction objective. Training unrolls the complete specialist update for K steps with the following normalized coefficient prediction loss

$$\mathcal{L}_r = \frac{1}{K} \sum_{k=1}^K \left[\frac{\|(\hat{\mathbf{a}}_{n+k}^{(r)} - \mathbf{a}_{n+k}^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}\|_2^2}{d_u^{(r)}} + \frac{\|(\hat{\mathbf{b}}_{n+k}^{(r)} - \mathbf{b}_{n+k}^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}\|_2^2}{d_p^{(r)}} \right]. \quad (16)$$

Hats denote recursively predicted coefficients, while unhatted coefficients are reference targets. The dimensions $d_u^{(r)}$ and $d_p^{(r)}$ are chart-specific; the scaling vectors $\boldsymbol{\sigma}_a^{(r)}$ and $\boldsymbol{\sigma}_b^{(r)}$ are estimated from the corresponding training set, and $\oslash$ denotes componentwise division. Progressively longer rollout horizons expose the learned corrections to their own accumulated prediction errors. The curricula and optimization settings are reported in Appendix D, and held-out trajectory splits are listed in Table 4.

This expression defines the rollout prediction term, not every term in every training configuration. Learned group-routing configurations also employ auxiliary group-balance, entropy, and Reynolds-partition supervision terms; their activation and coefficients are configuration-dependent.

3.4 CROSS-REGIME ROUTING AND FUSION

Unlike the internal routing in Section 3.2, the outer parameter-only regime router, denoted E2 in the experiments, produces a preference over the regime-local specialists:

$$\tilde{\mu}_R = (\mu - \mu_R)/\sigma_R, \quad \boldsymbol{\pi}(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu}_R)). \quad (17)$$

$\mathcal{R}_{E2}$ is a simple three-layer MLP—a 1–24–3 MLP with 123 parameters. It is trained on chart-assigned parameter labels, so $\boldsymbol{\pi}$ is a parameter-conditioned preference. The deterministic parameter-only output is identical for windows with the same μ and remains fixed during each rollout. The E2 normalization and training protocol are specified in Appendix D.

Prespecified adjacent candidates. The evaluation protocol supplies an ordered candidate pair $(r, s) = (S, H)$ or (P, H) on prespecified common chart support. For the supplied pair, define

$$\bar{\pi}_r = \frac{\pi_r}{\pi_r + \pi_s}, \quad \bar{\pi}_s = 1 - \bar{\pi}_r. \quad (18)$$

E2 Top-1 selects r when $\bar{\pi}_r \geq 1/2$ and s otherwise. The pair order means that the first-candidate weight always refers to Steady in the S-H study and Periodic in the H-P study.

History-conditioned output fusion. Both candidates encode the same physical initialization history in their own charts and advance independently. We refer to their history-conditioned physical-space fusion as T2-C (history-conditioned Top-2 fusion). Here, Top-2 denotes the two candidates in the prescribed adjacent pair, rather than the internally routed correction experts. T2-C adjusts the parameter-only pair preference using a chart-independent descriptor of the initial physical history:

$$\hat{\mathbf{q}}_{\text{T2C}}(t) = \alpha \hat{\mathbf{q}}^{(r)}(t) + (1 - \alpha) \hat{\mathbf{q}}^{(s)}(t), \quad \alpha = \text{sigmoid}(\text{logit}(\bar{\pi}_r^{\text{clip}}) + c_\psi([\tilde{\mu}_G; \mathbf{d}_n])). \quad (19)$$

Here $\bar{\pi}_r^{\text{clip}}$ is $\bar{\pi}_r$ clipped to $[10^{-6}, 1 - 10^{-6}]$, and $\mathbf{q}$ collects reconstructed velocity and gauge-corrected pressure on the common physical mesh. The eight descriptor channels $\mathbf{d}_n$ are computed from the initial three physical states, without future reference states. The gate input $[\tilde{\mu}_G; \mathbf{d}_n]$ uses training-window normalization, distinct from the trajectory-label normalization $\tilde{\mu}_R$ used by E2. The correction network is a 9-64-64-1 MLP with SiLU hidden activations and a zero-initialized final layer. Its construction and physical-field training loss are given in Appendix C.

The scalar α is computed for each prediction window and is shared by velocity and pressure throughout that window. Unlike E2, it may differ between histories at the same parameter. Fusion acts only on reconstructed outputs and is never fed back into either candidate. This interface does not require identical internal architectures for the two local predictors.

4 EXPERIMENTS

We organize the evidence into four questions: the accuracy and finite-horizon reliability of local specialists prediction, the benefit of cross-regime routing and fusion, the behavior of internal routing and physical-field diagnostics, and the computational cost. The three benchmarks provide complementary : flow pass a Circular case supplies controlled local comparisons, flow pass a Square case supplies the principal history-conditioned fusion study, and Fluidic Pinball case provides an archived POD-Galerkin comparison and additional overlap results. Their configurations and horizons are summarized in Table 4.

4.1 LOCAL PREDICTION ACCURACY AND FINITE-HORIZON RELIABILITY

Comparison with POD-Galerkin. Table 1 comparison local specialists with classical POD-Galerkin over $K = 32$ –56 steps. Table 1 show local specialists achieves low physical-field errors across all four evaluation settings. Compared with POD-Galerkin, the learned specialists consistently reduce both velocity and pressure errors, with the largest differences appearing in pressure. For example, the Hopf joint error decreases from 48.94% to 2.085%, while the standard Periodic error decreases from 43.20% to 0.8415%. The longer-horizon Periodic-core evaluation shows the same trend, with E_{joint} reduced from 41.91% to 1.672%. Together, these results show that the learned specialists maintain low physical-field error over the evaluated autonomous horizons. In particular, compared with POD-Galerkin, by incorporating a sparse mixture-of-experts correction into the projected Galerkin equation, the local specialists substantially reduce the long-horizon error.

Table 2: Circular local controls: full-window joint error $E_u + E_p$ (%) against POD-reconstructed reference fields. Neural entries are mean $\pm$ sample standard deviation over accepted seeds among three attempts. Accepted seed sets can differ between configurations; 2/3 rows are conditional descriptive statistics, not complete three-seed estimates. Repaired Hopf rows use FP32. OpInf is a single deterministic validation-selected fit. The two horizons use the same windows, with $K = 24$ restricted to their first 24 steps.

Regime	Model	$K = 24$	$K = 48$	Accepted
Hopf	Proposed, original	0.11521 ± 0.08694	0.11487 ± 0.10033	2/3
	Proposed, repaired	0.04334 ± 0.02379	0.05044 ± 0.03439	2/3
	Dense, active-matched	0.15406 ± 0.16364	0.15480 ± 0.15525	2/3
	Structured, original	0.12537 ± 0.04614	0.12724 ± 0.04302	3/3
	Structured, repaired	0.02229 ± 0.00432	0.02392 ± 0.00778	2/3
	OpInf	0.03870	0.05340	Deterministic
Periodic	Proposed, original	2.8791 ± 1.7269	4.9443 ± 3.1576	3/3
	Proposed, repaired	3.0667 ± 1.8373	5.5322 ± 3.6623	3/3
	Dense, active-matched	1.3251 ± 0.2268	2.3998 ± 0.6599	3/3
	Structured, original	2.8666 ± 1.4342	5.2810 ± 2.5679	3/3
	Structured, repaired	2.8598 ± 1.3081	5.0677 ± 2.3762	3/3
	OpInf	114.0941	114.9897	Deterministic

Table 1: Autonomous local-specialist prediction on the fluidic-pinball benchmark. Errors are physical-field percentages; lower is better. Periodic-core denotes the longer-horizon periodic evaluation; complete protocol details are given in Appendix E.2.

Evaluation setting	Model	E_u	E_p	E_{joint}
Steady ($K = 56$)	POD-Galerkin	—	—	—
	RAL-MoE-ROM	0.2247	0.7593	0.9840
Hopf ($K = 56$)	POD-Galerkin	0.4378	48.50	48.94
	RAL-MoE-ROM	0.0995	1.986	2.085
Periodic ($K = 32$)	POD-Galerkin	1.641	41.56	43.20
	RAL-MoE-ROM	0.2897	0.5518	0.8415
Periodic-core ($K = 56$)	POD-Galerkin	2.555	39.36	41.91
	RAL-MoE-ROM	0.6222	1.050	1.672

Controlled local comparisons. We compare Circular Hopf and Periodic predictors on common held-out windows at $K = 24$ and $K = 48$, using seeds 1248, 1600, and 2026 for the neural configurations. Dense matches the proposed model’s active neural parameter budget, while Structured retains explicit response branches but removes input-dependent expert routing. These are controls for the local architecture, not substitutes for a bare POD-Galerkin ablation. We also include a validation-selected operator inference (OpInf) fit as an external ROM comparison. The repaired configurations change the initialization of the low-rank quadratic factors; they are retrained models, not a post-hoc modification of the original predictions.

Table 2 brings together both regimes and horizons under the same full-window joint-error metric. Reference fields are reconstructed from true POD coefficients, so these errors exclude the original CFD projection remainder. Evaluation windows, aggregation, component errors, and implementation details are provided in Appendix E.1.

Accuracy under matched local controls. All Periodic neural configurations have three accepted seeds. Dense yields lower mean joint errors than the proposed configurations at both horizons, while Structured is close to the original proposed model. Under this active-parameter budget and evaluation protocol, the results therefore do not establish a Periodic accuracy advantage from input-dependent sparse routing. Errors increase when extending the window from 24 to 48 steps; neither low finite-window error nor successful completion proves arbitrary-time stability.

Initialization and training reliability. The original Circular quadratic branches remain inactive because both multiplicative factors were initialized at zero. Retaining one random factor while zeroing the other permits learning, but the repaired Proposed model does not improve the Periodic mean error. Retraining also changes optimization and checkpoint selection, so this comparison does not isolate the quadratic term alone. For Hopf, repaired FP32 configurations have only two accepted seeds per architecture, and their accepted sets differ from the original ones. Their lower conditional errors consequently do not establish improved expected performance or complete three-seed robustness. Appendix E.1 reports the initialization mechanism and Table 7 distinguishes training failure from validation rejection.

Comparison with operator inference. OpInf is competitive on Hopf at both horizons but substantially less accurate on Periodic in this task-specific fit. Its Periodic regularization choice lies at the upper boundary of the searched grid, limiting interpretation of the poor result. Together with the different pressure parameterization and derivative estimation described in Appendix E.1, this prevents a general claim of superiority over the OpInf method family.

4.2 CROSS-REGIME ROUTING AND PHYSICAL-SPACE FUSION

We hold the Square local predictors and E2 router fixed and compare assembly rules on the prescribed S–H and H–P support at $K = 24$. Each interface contains 16 held-out windows. E2 Top-1 selects within the supplied pair; T2-C learns a correction to its pair preference using initial-history descriptors. Neither the candidate set nor an automatic applicability mask is learned in this experiment.

Table 3 uses a common full-window metric. Train-fixed α and train-fixed logit offset are fitted only on training windows, and the fixed candidate is selected on validation error. These controls test whether the improvement can be explained by a constant mixture or calibration offset.

Table 3: Square full-window joint error (%) at $K = 24$ against the cached POD reference. Entries use the frozen-checkpoint reevaluation, not a three-seed average. Scalar controls use training data; the fixed candidate uses validation data.

Assembly rule	S–H	H–P
E2 Top-1	1.5786	1.6424
E2-probability blend	1.8145	1.4489
Equal blend	2.3849	1.3930
Train-fixed α	2.0591	0.9767
Train-fixed logit offset	2.0149	0.9821
Validation-fixed candidate	1.5786	1.0068
T2-C	1.0954	0.8123

T2-C improves on all listed controls for these fixed candidate sets. A separate architecture-matched μ -only control zeros the standardized history channels while retaining the gate architecture. Across three paired gate seeds, full-window joint errors are $1.095351 \pm 0.000015\%$ versus $1.814699 \pm 0.000242\%$ in S–H and $0.812343 \pm 0.000177\%$ versus $0.976939 \pm 0.000179\%$ in H–P. The reductions are 39.64% and 16.85%. These quantify gate-training variability, not uncertainty from retraining the complete hierarchy.

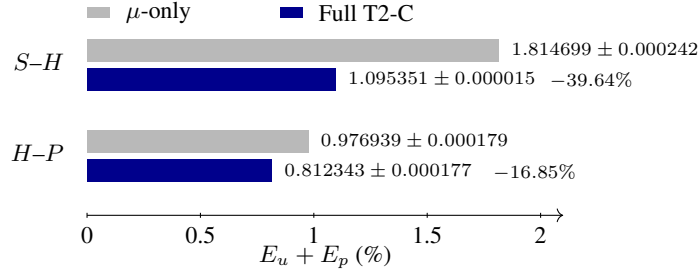


Figure 2: Full-window Square comparison over three paired gate seeds. All local candidates and E2 are fixed; bars show mean and sample standard deviation.

The evidence supports history-conditioned fusion for these windows, but does not prove that history dependence is necessary for every flow or gate. The weight is fixed within each prediction window; history can change it between windows. Appendix F.1 retains component errors, terminal-step results, scalar objectives, and complete field visualizations. Additional Pinball overlap results appear in Appendix F.2; their local candidates are Deep-FNN-H3, so they test the assembly interface rather than the sparse-MoE architecture. A posteriori convex oracles are diagnostics, not deployable methods.

4.3 INTERNAL ROUTING AND PHYSICAL-FIELD DIAGNOSTICS

Expert utilization. Post-hoc diagnostics use frozen Circular H/P specialists and are not used for checkpoint selection. Periodic distributes decisions across group-pair combinations; Hopf has a prescribed single group and fixed sparse support, but nonuniform, parameter-dependent mixture weights. Its single group is not evidence of collapse of a learned multi-group router. The detailed frequency, weight-mass, and entropy statistics in Appendix G.1 count the first integration stage per output step, not all RK evaluations. Utilization alone does not establish an accuracy benefit.

Fusion consistency. Appendix G.2 separates conditional algebraic identities from measured reconstructed-field diagnostics. On both Square interfaces, the measured T2-C pressure residual is lower than either candidate for both tested spatial stencils. However, Hopf has a smaller H-P momentum residual and a smaller S-H energy error. Fusion is therefore not a general residual or energy-error minimizer.

The measurements use offline least-squares spatial derivatives and coarse-time differences, not the original OpenFOAM finite-volume residuals. Nonzero reference residuals include POD truncation and differentiation error. Identity discrepancies near machine precision verify the algebra, not conservation or physical accuracy. The phase screen resolves none of the 16 S-H windows and 15 of 16 H-P windows; even resolved short-window traces do not establish absence of aliasing or long-time phase stability. The complete energy, residual, probe, and phase curves are retained in Appendix G.2.1.

4.4 COMPUTATIONAL COST AND TIMING BOUNDARIES

Sparse activation is a parameter-use property, not a runtime claim. Circular H/P activate 44.53% and 17.09% of their trainable parameters per analyzed prediction decision. For native Circular Periodic prediction at batch size one and $K = 48$, median times are 2.1848 s (proposed), 0.6579 s (Dense), and 0.8652 s (Structured). The proposed implementation is therefore slower than Dense in this measurement despite sparse activation; the native Galerkin path also executes on the CPU.

A separate warm physical-input/physical-output measurement includes history encoding, local advancement, and all-step field decoding. It takes 2.18747 s on an RTX 4090 with a Xeon host for a Circular Periodic window spanning 598.70093 physical time units. A matched historical production FOM interval takes 1996 s. The resulting ratio of approximately 912.5 is not a full-hierarchy or hardware-independent speedup: hardware and I/O boundaries differ, the FOM timing has one realization, and ROM loading, output writes, E2, descriptors, and T2-C are excluded. Appendix H

lists warm-ups, repetitions, synchronization, memory, hardware, and all included/excluded work. A fully paired Top-2 end-to-end timing remains unmeasured.

5 CONCLUSION

We introduced RAL-MoE-ROM to combine autonomous regime-local predictors through physical-space assembly while retaining independent reduced coordinates. On the evaluated centered-square overlaps, physical-history conditioning improves T2-C over the architecture-matched parameter-only gate for all three gate-training seeds. The local control experiments qualify the role of the inner architecture: the active-parameter-matched Dense control is more accurate on Circular Periodic, and the original audited quadratic branches remained inactive. Repaired Hopf training also retains failed seeds. These results support the evaluated local prediction and output-assembly capabilities, but do not establish sparse routing or explicit quadratic corrections as necessary sources of accuracy. The study assumes prescribed regime coverage and supplied initial physical histories. Full-field CFD errors, nonlinear constraint preservation, global cross-chart data-role verification, and end-to-end FOM speedups remain to be established.

REPRODUCIBILITY STATEMENT

The main text reports the benchmark roles, evaluation horizons, baselines, and metrics. Appendices A–D document the problem, projected operators, specialist architecture, routing, and implementation. Experimental appendices follow the main-text order: local prediction (Appendix E), fusion (Appendix F), routing and physical-field diagnostics (Appendix G), and cost (Appendix H). Appendix G.1 reports post-hoc internal-routing statistics from fixed validation-selected Hopf and Periodic specialists on held-out rollout windows; these diagnostics are not used for model selection. Appendix E.1 reports the new Circular control study, initialization audit, and accepted/failed seed counts; Appendix H separates parameter counts from measured local latency. Appendix F.1 additionally reports an architecture-matched descriptor ablation and three-seed variability of the centered-square T2-C gate, with the underlying specialists and E2 router held fixed.

AI USE STATEMENT

Generative AI tools, including ChatGPT and Codex, assisted with language polishing, LaTeX formatting, consistency checks, experiment-code development, and analysis of experimental records. The authors take full responsibility for reviewing and verifying all scientific content, mathematical formulations, experimental configurations, numerical results, and conclusions, and for the final manuscript and all reported results.

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A FULL PROBLEM SETUP AND WEIGHTED POD CONSTRUCTION

A.1 PARAMETERIZED FULL-ORDER MODEL, GAUGE, AND REGIME COVERAGE

Finite-volume spatial discretization of the Navier–Stokes equations in Eq. (1) yields a semi-discrete system, as in Eq. (2), governing the velocity $\mathbf{u}(t; \mu) \in \mathbb{R}^{N_u}$ and pressure $p(t; \mu) \in \mathbb{R}^{N_p}$:

$$\dot{\mathbf{u}} = \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu)\mathbf{u} - \mathcal{N}_h(\mathbf{u}, \mathbf{u}; \mu) - \nabla_h(\mu)p, \quad \text{div}_h(\mu)\mathbf{u} = 0. \quad (20)$$

Here $\mathcal{L}_h$ and $\mathcal{N}_h$ denote the spatially discretized viscous and convective operators, while ∇_h and div_h denote the pressure gradient and velocity divergence operators. The dot denotes a continuous-time derivative; time advancement of the reduced model is specified in Section 3.2. Pressure is defined only up to an additive constant. With positive finite-volume weights $\mathbf{w}$, each pressure snapshot is mapped to the weighted zero-mean gauge

$$\Gamma_p(q) = q - \frac{\mathbf{w}^\top q}{\mathbf{w}^\top \mathbf{1}} \mathbf{1}, \quad p^\circ = \Gamma_p(p), \quad \mathbf{w}^\top p^\circ = 0. \quad (21)$$

The parameter domain is covered by overlapping regime-local parameter regions,

$$\mathcal{M} \subseteq \bigcup_{r \in \{S, H, P\}} \mathcal{M}_r, \quad \mathcal{O}_{SH} = \mathcal{M}_S \cap \mathcal{M}_H, \quad \mathcal{O}_{HP} = \mathcal{M}_H \cap \mathcal{M}_P, \quad (22)$$

where S , H , and P denote the steady, Hopf-onset, and periodic regimes, respectively. Only adjacent overlaps are used; no direct S – P overlap is considered.

A.2 AREA-WEIGHTED LOCAL POD

The finite-volume quadrature induces the weighted inner products

$$\langle \mathbf{v}, \mathbf{z} \rangle_{M_u} = \mathbf{v}^\top M_u \mathbf{z}, \quad \langle q, s \rangle_{M_p} = q^\top M_p s, \quad M_p = \text{diag}(\mathbf{w}), \quad M_u = \text{diag}(M_p, M_p). \quad (23)$$

For chart r , let $d_u^{(r)}$ and $d_p^{(r)}$ denote the retained velocity and pressure dimensions. Let $X_u^{(r)}$ and $X_p^{(r)}$ collect the centered velocity and gauge-corrected pressure snapshots from the training trajectories assigned to that chart. The corresponding chart-specific means are denoted by $\bar{\mathbf{u}}^{(r)}$ and $\bar{p}^{(r), \circ}$. The local POD bases solve

$$\Phi_u^{(r)} \in \arg \min_{\Psi^\top M_u \Psi = I_{d_u^{(r)}}} \left\| X_u^{(r)} - \Psi \Psi^\top M_u X_u^{(r)} \right\|_{M_u, F}^2, \quad (24)$$

$$\Phi_p^{(r)} \in \arg \min_{\Psi^\top M_p \Psi = I_{d_p^{(r)}}} \left\| X_p^{(r)} - \Psi \Psi^\top M_p X_p^{(r)} \right\|_{M_p, F}^2, \quad (25)$$

where

$$\|Y\|_{M, F}^2 = \text{tr}(Y^\top M Y). \quad (26)$$

Equivalently, consider the weighted singular value decompositions

$$M_u^{1/2} X_u^{(r)} = U_u^{(r)} \Sigma_u^{(r)} V_u^{(r)\top}, \quad M_p^{1/2} X_p^{(r)} = U_p^{(r)} \Sigma_p^{(r)} V_p^{(r)\top}. \quad (27)$$

The retained bases are

$$\Phi_u^{(r)} = M_u^{-1/2} U_u^{(r)}(:, 1:d_u^{(r)}), \quad \Phi_p^{(r)} = M_p^{-1/2} U_p^{(r)}(:, 1:d_p^{(r)}), \quad (28)$$

and satisfy

$$\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I, \quad \Phi_p^{(r)\top} M_p \Phi_p^{(r)} = I. \quad (29)$$

Because the pressure snapshots and their chart mean satisfy the same weighted gauge condition, the pressure basis also satisfies

$$\mathbf{w}^\top \Phi_p^{(r)} = \mathbf{0}^\top. \quad (30)$$

Encoding and decoding are performed independently within each chart:

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \mathcal{D}_r^u(\mathbf{a}^{(r)}) = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (31)$$

$$\mathbf{b}^{(r)} = \Phi_p^{(r)\top} M_p (\Gamma_p(p) - \bar{p}^{(r),\circ}), \quad \mathcal{D}_r^p(\mathbf{b}^{(r)}) = \bar{p}^{(r),\circ} + \Phi_p^{(r)} \mathbf{b}^{(r)}. \quad (32)$$

The reduced coefficients are standardized using chart-local statistics estimated from the corresponding training trajectories:

$$\tilde{\mathbf{a}}^{(r)} = (\mathbf{a}^{(r)} - \boldsymbol{\mu}_a^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}, \quad \tilde{\mathbf{b}}^{(r)} = (\mathbf{b}^{(r)} - \boldsymbol{\mu}_b^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}. \quad (33)$$

Additional specialist features and their normalization are specified in Section 3.2.

B PROJECTED GALERKIN AND PRESSURE OPERATORS

B.1 CHART-LOCAL GALERKIN MOMENTUM DYNAMICS

For each chart $r \in \{S, H, P\}$, substituting the chart-local affine reconstructions from Appendix A into the semi-discrete momentum equation (20) and testing against the local velocity basis in the M_u inner product yields

$$\dot{\mathbf{a}}^{(r)} = \mathbf{c}_r(\mu) + A_r(\mu) \mathbf{a}^{(r)} + H_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + C_r^p(\mu) \mathbf{b}^{(r)}. \quad (34)$$

Here “:” denotes the double contraction. For velocity modes $\phi_i^{u,(r)}$, $i = 1, \dots, d_u^{(r)}$, and pressure modes $\phi_\ell^{p,(r)}$, $\ell = 1, \dots, d_p^{(r)}$, the projected operators are

$$[\mathbf{c}_r(\mu)]_i = \left\langle \phi_i^{u,(r)}, \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu) \bar{\mathbf{u}}^{(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \bar{\mathbf{u}}^{(r)}; \mu) - \nabla_h(\mu) \bar{p}^{(r),\circ} \right\rangle_{M_u}, \quad (35)$$

$$[A_r(\mu)]_{ij} = \left\langle \phi_i^{u,(r)}, \mathcal{L}_h(\mu) \phi_j^{u,(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \phi_j^{u,(r)}; \mu) - \mathcal{N}_h(\phi_j^{u,(r)}, \bar{\mathbf{u}}^{(r)}; \mu) \right\rangle_{M_u}, \quad (36)$$

$$[H_r(\mu)]_{ijk} = -\left\langle \phi_i^{u,(r)}, \mathcal{N}_h(\phi_j^{u,(r)}, \phi_k^{u,(r)}; \mu) \right\rangle_{M_u}, \quad (37)$$

$$[C_r^p(\mu)]_{i\ell} = -\left\langle \phi_i^{u,(r)}, \nabla_h(\mu) \phi_\ell^{p,(r)} \right\rangle_{M_u}. \quad (38)$$

We denote the resulting chart-local projected momentum vector field by

$$\mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) = \mathbf{c}_r(\mu) + A_r(\mu) \mathbf{a}^{(r)} + H_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + C_r^p(\mu) \mathbf{b}^{(r)}. \quad (39)$$

This is the projected Galerkin equation summarized in Eq. (4); the additive hybrid specialist augments it with the learned closure in Eq. (5).

B.2 REDUCED PRESSURE-POISSON EQUATION

Taking the divergence of the incompressible momentum equation with spatially constant viscosity gives

$$-\Delta p^\circ = \nabla \cdot [(\mathbf{u} \cdot \nabla) \mathbf{u} - \mathbf{f}], \quad \int_{\Omega} p^\circ d\Omega = 0, \quad (40)$$

where p° denotes gauge-corrected pressure. Substituting equation 32 into equation 40, and applying local pressure-mode testing, integration by parts, and Appendix A quadrature for the volume integrals, yields a reduced pressure system.

$$K_r^p(\mu) \mathbf{b}^{PP,(r)} = \mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a}^{(r)} + Q_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}), \quad (41)$$

where

$$[K_r^p]_{\ell m} = -\left\langle \nabla_h \phi_\ell^{p,(r)}, \nabla_h \phi_m^{p,(r)} \right\rangle_{M_u}, \quad (42)$$

$$[\mathbf{d}_r]_\ell = \left\langle \nabla_h \phi_\ell^{p,(r)}, \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \bar{\mathbf{u}}^{(r)}; \mu) - \mathbf{f}_h(\mu) + \nabla_h \bar{p}^{(r),\circ} \right\rangle_{M_u}, \quad (43)$$

$$[B_r]_{\ell j} = \left\langle \nabla_h \phi_\ell^{p,(r)}, \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \phi_j^{u,(r)}; \mu) + \mathcal{N}_h(\phi_j^{u,(r)}, \bar{\mathbf{u}}^{(r)}; \mu) \right\rangle_{M_u}, \quad (44)$$

$$[Q_r]_{\ell jk} = \left\langle \nabla_h \phi_\ell^{p,(r)}, \mathcal{N}_h(\phi_j^{u,(r)}, \phi_k^{u,(r)}; \mu) \right\rangle_{M_u}. \quad (45)$$

The resulting algebraic pressure map is

$$\mathcal{Q}_r^{PP}(\mathbf{a}^{(r)}; \mu) = (K_r^p(\mu))^\dagger \left[\mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a}^{(r)} + Q_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) \right], \quad (46)$$

where † denotes the Moore–Penrose pseudoinverse and coincides with the inverse when K_r^p is non-singular. The pressure field is then reconstructed as $\mathcal{Q}_r^{PP}(\mathbf{a}^{(r)}; \mu)$. Thus, pressure is recovered algebraically from the local velocity state rather than advanced as an independent dynamical variable.

C OUTER ROUTING AND T2-C DETAILS

Chart-independent physical-history descriptor. The implementation in Section 3.4 uses eight descriptor channels and one parameter input. These features are computed from the initial three physical states at $t_{n-2} < t_{n-1} < t_n$, with pressure mapped to the common zero-mean gauge. They are independent of the specialists’ POD coordinates. All quantities below use the nondimensional data conventions of the experiment, and $\epsilon = 10^{-12}$ is the numerical floor.

Define $A_\Omega = \sum_i w_i$, $U_j = \|\mathbf{u}_j\|_{M_u}^2$, $P_j = \|p_j^\circ\|_{M_p}^2$, and

$$h_j = \max(t_j - t_{j-1}, \epsilon), \quad s_j^u = \sqrt{\max(U_j, \epsilon)}, \quad s_j^p = \sqrt{\max(P_j, \epsilon)}, \quad (47)$$

$$v_j = (\mathbf{u}_j - \mathbf{u}_{j-1})/h_j, \quad g_j = \frac{\log \max(U_j, \epsilon) - \log \max(U_{j-1}, \epsilon)}{h_j}. \quad (48)$$

The ordered raw descriptor is

$$\mathbf{s}_n = \text{col} \begin{pmatrix} \frac{1}{2} \log \max(U_n/A_\Omega, \epsilon), \\ \frac{1}{2} \log \max(P_n/A_\Omega, \epsilon), \\ \log(1 + \|v_n\|_{M_u}/(s_n^u + \epsilon)), \\ \log(1 + \|v_{n-1}\|_{M_u}/(s_{n-1}^u + \epsilon)), \\ \tanh(g_n), \\ \tanh(g_n - g_{n-1}), \\ \log(1 + \|p_n^\circ - p_{n-1}^\circ\|_{M_p}/[h_n(s_n^p + \epsilon)]), \\ \log(1 + \|v_n - v_{n-1}\|_{M_u}/(s_n^u + \epsilon)) \end{pmatrix}. \quad (49)$$

The last channel measures a difference of velocity rates: it is not divided again by a time interval and is not the second-derivative descriptor in an acceleration discretization. The log and hyperbolic-tangent transformations above are part of the actual feature construction. For the gate, the nine raw features $[\mu; \mathbf{s}_n]$ are standardized using the means and population standard deviations of training windows only. Any feature standard deviation below 10^{-8} is replaced by one. We denote the resulting vector by $[\tilde{\mu}_G; \mathbf{d}_n]$; its parameter normalization is fitted for the gate and is distinct from the E2 training-row normalization $\tilde{\mu}_R$. The architecture-matched μ -only control sets all eight descriptor channels to zero after standardization.

Physical-space T2-C fusion. The logit-correction network receives $[\tilde{\mu}_G; \mathbf{d}_n] \in \mathbb{R}^9$ and uses two hidden layers of width 64 with SiLU activations followed by a scalar linear output (9–64–64–1; 4,865 trainable parameters). The final layer is initialized at zero, so the initial correction vanishes and the fusion weight reduces to the E2 pair preference. Specifically,

$$\Delta_\psi = c_\psi([\tilde{\mu}_G; \mathbf{d}_n]), \quad \alpha = \text{sigmoid}(\ell^{E2} + \Delta_\psi). \quad (50)$$

The two selected specialists encode the same physical history in their respective local charts and then evolve independently. The scalar α is fixed throughout the prediction window and is shared by velocity and pressure. After reconstruction on the common physical mesh, T2-C forms

$$\hat{\mathbf{q}}_\alpha^c(t) = \alpha \hat{\mathbf{q}}_1^c(t) + (1 - \alpha) \hat{\mathbf{q}}_2^c(t), \quad c \in \{u, p\}, \quad (51)$$

where the pressure field is gauge corrected. The fused field is an output only and is never projected back into either specialist.

D REPRODUCIBILITY AND IMPLEMENTATION DETAILS

Physical-field error metrics. We evaluate reconstructed velocity and gauge-corrected pressure using the relative weighted field errors

$$E_u(n) = 100 \frac{\|\hat{\mathbf{u}}_n - \mathbf{u}_n\|_{M_u}}{\|\mathbf{u}_n\|_{M_u}}, \quad E_p(n) = 100 \frac{\|\hat{p}_n^\circ - p_n^\circ\|_{M_p}}{\|p_n^\circ\|_{M_p}}. \quad (52)$$

Here $\|v\|_M = (\mathbf{v}^\top M \mathbf{v})^{1/2}$ uses the finite-volume quadrature introduced in Appendix A; the velocity norm includes both components, and p° denotes pressure in the common zero-mean gauge. Unless stated otherwise, benchmark-specific temporal and trajectory aggregation is reported with the corresponding experiment. Whenever a joint score is reported, $E_{\text{joint}} = E_u + E_p$.

Trajectory-level partitioning. Table 4 summarizes the resulting chart-wise parameter coverage, reduced dimensions, trajectory splits, and evaluation horizons. All splits are defined at the level of complete parameter trajectories rather than individual snapshots or rollout windows. Local bases, normalization statistics, and model parameters are constructed from training trajectories only.

Table 4: Chart-wise parameter coverage, reduced dimensions, trajectory splits, and evaluation horizons. K denotes the number of autonomous rollout steps used for the corresponding local evaluation.

Benchmark	Chart	Re range	(r_u, r_p)	Train	Val.	Test	K
Circular cylinder	S	20.0–46.4	(32, 32)	14	2	4	56
	H	45.5–59.2	(32, 32)	29	2	3	56
	P	57.3–200	(32, 32)	53	6	4	48
Centered square	S	50–95.4	(5, 4)	60	5	4	16
	H	94–102	(11, 11)	23	6	5	48
	P	98.5–150	(28, 26)	27	6	4	24
Fluidic pinball	S	10–19	(4, 3)	37	8	7	56
	H	16.6–25	(5, 5)	39	8	7	56
	P	21.25–60	(17, 16)	47	9	9	32

T2-C gate optimization. Both full T2-C and the architecture-matched μ -only gate use AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), 8,000 steps, and gradient-norm clipping at 1. Mini-batches sample training windows with replacement. Validation is evaluated at step 1 and every 100 steps; checkpoint selection minimizes the largest Reynolds-number-wise full-window mean joint error. The evaluation is reported in Appendix F.1.

Benchmark discretization and sampling. The centered-square dataset contains 120 complete trajectories over $Re \in [50, 150]$ on a 9,400-cell finite-volume mesh. The CFD time step is $\delta t_{\text{CFD}} = 0.02$, and snapshots are stored every 200 CFD steps, giving $\Delta t_{\text{snap}} = 4$. The fluidic-pinball dataset contains 131 complete trajectories over $Re \in [10, 60]$ on a 70,194-cell mesh, with $\delta t_{\text{CFD}} = 0.005$ and a snapshot interval of 0.25 physical time units. The circular-cylinder dataset contains 100 Reynolds-parameterized trajectories spanning $Re \in [20, 200]$ on a common finite-volume mesh represented by 97,368 spatial points. Trajectories contain different numbers of snapshots, yielding 14,167 snapshots in total. The unsteady simulations employ adaptive CFD time stepping subject to $Co_{\text{max}} = 0.8$ and $\delta t_{\text{max}} = 0.2$, while the snapshot interval is adjusted according to the estimated vortex-shedding period, with eight snapshots retained per shedding cycle. The chart-wise parameter ranges, reduced dimensions, splits, and evaluation horizons are summarized in Table 4.

Specialist optimization. Table 5 summarizes the reported training configurations, which use AdamW with chart-specific learning rates, weight decay, and rollout curricula.

Table 5: Reported specialist training configurations. “Selected” denotes the validation-selected checkpoint. The curriculum and budget describe the configured schedule, which may extend beyond the selected checkpoint.

Specialist	Initial lr	Weight decay	Rollout curriculum	Budget	Selected
Circular S	1.55×10^{-5}	1.0×10^{-4}	4,8,16	3600 steps	1200 steps
Circular H	1.0×10^{-3}	1.0×10^{-4}	1,2,4,8	8000 steps	7200 steps
Circular P	5.5×10^{-4}	1.5×10^{-4}	4,8,12,16	240 epochs	85 epochs
Square S	1.55×10^{-5}	1.0×10^{-4}	1,4,8,16	4000 steps	3200 steps
Square H	1.0×10^{-3}	1.0×10^{-4}	1,2,4,8	8000 steps	7200 steps
Square P	5.5×10^{-4}	1.5×10^{-4}	4,8,12,16	720 epochs	490 epochs
Pinball S	1.0×10^{-3}	1.0×10^{-4}	4,8,16	8000 steps	400 steps
Pinball H	1.0×10^{-3}	1.0×10^{-4}	4,8,16,32,56	8000 steps	8000 steps
Pinball P	1.0×10^{-3}	1.0×10^{-4}	4,8,16,24,32	8000 steps	8000 steps

Outer E2 router configuration. The parameter-only router is an MLP $1 \rightarrow 24 \rightarrow 3$ with a Tanh hidden activation and 123 trainable parameters. The scalar Reynolds number is standardized using training-row mean and population standard deviation; a standard deviation below 10^{-7} is replaced by one. Within each chart and split, each unique Reynolds number supplies one row with that chart label. A parameter occurring in more than one chart can therefore produce multiple labeled rows, rather than an assumed unique bifurcation label. The loss is class-weighted cross entropy plus $0.05 \text{ mean}(\pi_S \pi_P)$, with class weight $N/(3N_c)$. Training uses AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), a cosine schedule over 8,000 steps ending at 1.5×10^{-5} , sampling with replacement in batches of $\min(256, N)$, and gradient-norm clipping at one. Selection maximizes validation balanced accuracy plus macro-F1 minus 10^{-3} times the current training-batch loss, rather than validation cross entropy alone.

Specialist architectures. For sparse MoE, $D = 13 + 5(2d_u + d_p)$ is feature width. Features include parameter terms, current coefficients and projected Galerkin equation, 11 derived norm/energy/ratio quantities, and two historical blocks. The time step is an integrator input, not a network feature. Each encoder has three linear layers of width h with LayerNorm and SiLU; the first two include dropout 0.04. Two refinement residual blocks follow. Velocity and pressure share this encoder, but not their router or expert parameters.

Each group contains six routed experts and one shared expert per channel, with separate velocity/pressure Top-2 supports. Routers receive $[z; \xi]$ and use LayerNorm, a width- h SiLU hidden layer, dropout 0.04, and a linear logit head. Each expert projects $[z; s^c]$ to width h , applies B expanded residual feed-forward blocks with GELU, and maps to the channel output. Residual scales are learned scalars initialized to 0.1. The explicit linear branch has no bias; the two quadratic factors each have shape $d_c \times 4 \times \dim(s^c)$, with $\kappa = 0.05$ and no additional output projection. The branches are added within each expert before sparse mixture weighting. Outputs are converted from standardized target coordinates before combination with physical-model terms.

E LOCAL PREDICTION: PROTOCOLS, CONTROLS, AND ROBUSTNESS

E.1 CIRCULAR-CYLINDER CONTROLLED REEVALUATION

Evaluation and aggregation protocol. The neural study uses seeds 1248, 1600, and 2026. Hopf has 42 windows from $Re = 47.081355, 49.022357, 51.786450$; Periodic has 32 windows from $Re = 70.314635, 100.352251, 149.059229, 189.862278$. Each window contains 48 prediction steps, with $K = 24$ using its first 24 steps. Relative physical-field errors are averaged over steps, then windows within each Reynolds number, then equally over Reynolds numbers. Means and sample standard deviations are computed over accepted seeds. Reference fields are reconstructed from true POD coefficients on the corresponding basis. Only checkpoints accepted by the original validation rule are evaluated on held-out windows. Table 5 describes the archived original training configurations, not the selected checkpoints of every repeated run.

Table 6: Circular Periodic component errors (%). Neural entries are mean $\pm$ sample standard deviation over three accepted seeds. OpInf is deterministic. Joint errors are reported in the main-text comparison and are not repeated here. Components and joint values are independently rounded.

Model	$K = 24$		$K = 48$	
	E_u	E_p	E_u	E_p
Proposed, original	0.5731 ± 0.3638	2.3060 ± 1.3681	1.0107 ± 0.6513	3.9337 ± 2.5070
Proposed, repaired	0.5814 ± 0.3949	2.4854 ± 1.4427	1.1040 ± 0.7840	4.4282 ± 2.8786
Dense, active-matched	0.2446 ± 0.0323	1.0805 ± 0.1946	0.4790 ± 0.1208	1.9208 ± 0.5391
Structured, original	0.5178 ± 0.2667	2.3489 ± 1.1688	1.0405 ± 0.5015	4.2406 ± 2.0664
Structured, repaired	0.5429 ± 0.2783	2.3168 ± 1.0315	0.9983 ± 0.4944	4.0693 ± 1.8827
OpInf	15.8676	98.2264	17.6639	97.3258

Table 7: Hopf completion and validation acceptance over three attempted seeds per architecture. AMP and FP32 are separate repaired campaigns, not pooled repeats.

Configuration	Proposed	Structured	Failure or rejection
Original	2/3	3/3	Proposed seed 1600 not accepted by validation
Repaired, AMP	0/3	0/3	Non-finite gradients
Repaired, FP32	2/3	2/3	Proposed 2026: step 3220; Structured 1248: step 2860, both during $K = 4$

Dense matches active neural parameter count, not total stored parameters or execution time. Structured removes input-dependent expert routing while retaining explicit response branches. Neither is identical to the older Vanilla-FNN-MoE or Global MoE configuration. The present protocol does not retroactively validate the earlier Circular min-max summaries or establish the identity of the unresolved Steady comparison. Joint errors and their interpretation are reported once in Table 2 and the accompanying main-text discussion; the supplementary table below supplies component errors.

Quadratic-factor initialization. For a multiplicative response $(v_1^\top s)(v_2^\top s)$, the gradient of either factor contains the other factor. Initializing both at zero therefore prevents learning in this branch. Each accepted original Hopf Proposed checkpoint contains 28 quadratic-factor tensors, all zero. Each original Periodic Proposed checkpoint contains 84, and each Periodic Structured checkpoint contains four, also all zero. The repaired initialization preserves a random factor and zeros only its partner. Smoke training updates all 14 factor pairs for Hopf Proposed, 42 for Periodic Proposed, and two for Periodic Structured. This check measures branch learning activity, separately from the accuracy comparison in the main text.

Training completion and validation acceptance. Table 7 records the attempted Hopf campaigns. Acceptance requires a usable run and the prescribed validation rule; a missing accepted checkpoint is not automatically a divergent held-out rollout. The original Dense control also has two accepted seeds out of three.

The FP32 preflight covers 116 combinations of 29 training parameters and four curriculum lengths. Passing it does not guarantee finite gradients later in optimization. The repaired FP32 training-failure rate is 1/3 per architecture (2/6 pooled across these two configurations). The earlier AMP campaign failed for all six attempts and is not pooled with FP32. Original validation rejection is distinct from non-finite-gradient training failure. Failed seeds are not replaced by post-hoc retries in the reported statistics.

OpInf adaptation and validation. The regularized polynomial model uses constant, linear, compact quadratic, standardized $1/Re$, and parameter-state interaction features. Training-trajectory derivatives are estimated by second-order numerical differentiation. Ridge regularization is selected on validation windows, with a non-finite window invalidating a candidate. Prediction uses fixed RK4 with four substeps per output interval. Pressure is an algebraic ridge output using the same polynomial features rather than the proposed PPE-plus-correction map. Test windows, reference

reconstructions, and quadrature match the neural comparison. The upper endpoint of the Periodic ridge search is 100. This is a task-specific adaptation, not a reproduction of every implementation detail of an external method.

E.2 ARCHIVED PINBALL LOCAL PREDICTIONS

The POD–DeepONet baseline provides a complementary comparison between one-step prediction and autonomous rollout. It attains $E_u = 1.5921\%$ and $E_p = 23.3447\%$ on 4,228 held-out input–output pairs, but only 1 of 28 held-out trajectories completes the prescribed autonomous rollout evaluation. This difference illustrates that one-step accuracy does not by itself imply reliable autonomous prediction. Because the POD–DeepONet evaluation uses a different temporal cadence from the Periodic specialist, it is used only for this comparison of one-step prediction and autonomous rollout and not as a cadence-matched long-horizon accuracy baseline.

Periodic-core evaluation protocol. The $K = 32$ Periodic result in Table 1 corresponds to the standard periodic evaluation. We additionally report a longer $K = 56$ evaluation restricted to the Periodic-core final-test trajectories. The full-order snapshots are stored every 0.25 physical time units; subsampling every fourth snapshot gives an effective interval $\Delta t = 1$, matching the Periodic specialist’s native history cadence. The $K = 56$ setting therefore provides a longer autonomous evaluation within the same periodic regime without changing the specialist representation or sampling convention.

Predictor identity and interpretation. The Periodic autonomous values are associated with Deep-FNN-H3. Formal Pinball outer-candidate seals identify all three local candidates as Deep-FNN-H3, but do not resolve every earlier standalone Steady/Hopf table row. Table 1 is retained as an archived comparison, not a matched ablation of adding sparse MoE to bare Galerkin. Its finite evaluation lengths do not prove arbitrary-time stability. No baseline Steady error is inferred from the missing entry.

F ROUTING AND FUSION: COMPLETE RESULTS

F.1 CENTERED-SQUARE RESULTS

This appendix provides the complete centered-square overlap results underlying the main-text routing and fusion comparison. Table 8 reports the velocity, pressure, and joint errors for the individual specialists and the fusion baselines in the S – H and H – P overlap regions. In the main text, “Best fixed” denotes the fixed specialist with the lowest aggregate joint error over the corresponding overlap, whereas the convex oracle is a window-wise a posteriori reference.

Overlap evaluation and descriptor ablation. For both overlaps, E_u , E_p , and $E_{\text{joint}} = E_u + E_p$ are reported at the terminal rollout step ($k = 24$) and averaged over 16 held-out windows. This terminal metric differs from the full-window objective used by the diagnostic convex oracle. To assess the role of history information in the second routing level, we also include T2-C μ -only, which uses the same correction-gate architecture and training protocol as T2-C but removes the standardized history descriptor by setting its channels to zero. Table 8 reports the validation-selected comparison, and the corresponding three-seed gate variability is summarized separately below.

Table 8: Centered-square routing and fusion errors at $K = 24$. The convex oracle is an a posteriori reference unavailable during prediction. T2-C μ -only uses the same gate architecture as T2-C but removes the standardized history descriptor. Bold indicates the best deployable method within each overlap.

Overlap	Predictor	E_u	E_p	E_{joint}
$S-H$	Steady specialist	2.5976	2.2450	4.8426
	Hopf specialist	0.4942	1.2356	1.7298
	E2 Top-1	0.4942	1.2356	1.7298
	Equal blend	1.3644	1.6677	3.0321
	E2-probability blend	0.7898	1.3949	2.1847
	T2-C μ -only	0.7898	1.3948	2.1846
	T2-C	0.4615	0.7343	1.1958
	Convex oracle	0.4659	0.7183	1.1842
$H-P$	Periodic specialist	0.6420	0.5659	1.2079
	Hopf specialist	1.0034	1.8684	2.8718
	E2 Top-1	0.7422	1.1052	1.8473
	Equal blend	0.6077	1.0069	1.6146
	E2-probability blend	0.6154	1.0464	1.6618
	T2-C μ -only	0.5875	0.5629	1.1504
	T2-C	0.4509	0.3956	0.8465
	Convex oracle	0.4479	0.3905	0.8384

Table 9: Parameter-wise centered-square routing and T2-C results at $K = 24$. Joint errors and relative reductions are reported in percent. The mean weight is the Steady-specialist weight for $S-H$ and the Periodic-specialist weight for $H-P$.

Overlap	Re	E2 choice	$\bar{\alpha}$	E2 E_{joint}	T2-C E_{joint}	Reduction
$S-H$	95.1	Hopf	0.0967	1.7140	1.1827	31.00%
$S-H$	95.3	Hopf	0.0971	1.7456	1.2090	30.74%
$H-P$	100.5	Hopf	0.5870	2.6485	0.8571	67.64%
$H-P$	102.0	Periodic	0.7094	1.0462	0.8360	20.09%

The regime-local specialists also exhibit low aggregate physical-field errors under their respective held-out evaluations: $(E_u, E_p) = (1.5311, 1.4583)\%$ in the Steady regime, $(0.4387, 0.4844)\%$ in the Hopf regime, and $(0.9395, 1.0921)\%$ in the Periodic regime. Within the overlap evaluations, however, the more accurate individual specialist differs by interface: the Hopf specialist is the better single-model reference in $S-H$, whereas the Periodic specialist is more accurate in $H-P$. This overlap dependence motivates an adaptive assembly rule rather than a fixed specialist choice across neighboring regimes.

Gate-training seed variability. To assess the sensitivity of the second routing level, we retrain only the T2-C gate for seeds 42001, 42002, and 42003, keeping the specialists, E2 router, candidate rollouts, data splits, and normalization fixed. Full T2-C and the architecture-matched μ -only variant share the optimization and selection protocol in Appendix D. Table 10 reports the mean and sample standard deviation ($N = 3$, denominator $N - 1$), and Table 11 lists the seed-wise joint errors. These statistics describe gate-training variability, not across- Re variation, a confidence interval, or full-pipeline uncertainty.

Table 10: Three-seed variability of the centered-square T2-C gate at the $K = 24$ terminal step. Values are reported as mean $\pm$ sample standard deviation across gate-training seeds.

Overlap	Gate	E_u	E_p	E_{joint}
$S-H$	T2-C μ -only	0.79006 ± 0.00026	1.39498 ± 0.00012	2.18504 ± 0.00038
	Full T2-C	0.46156 ± 0.00004	0.73416 ± 0.00014	1.19572 ± 0.00010
$H-P$	T2-C μ -only	0.59067 ± 0.00461	0.56050 ± 0.00435	1.15116 ± 0.00070
	Full T2-C	0.45211 ± 0.00138	0.39793 ± 0.00215	0.85004 ± 0.00347

Table 11: Seed-wise centered-square terminal joint errors E_{joint} (%) at $K = 24$.

Overlap	Gate	42001	42002	42003
$S-H$	T2-C μ -only	2.18460	2.18524	2.18527
	Full T2-C	1.19584	1.19567	1.19565
$H-P$	T2-C μ -only	1.15037	1.15170	1.15143
	Full T2-C	0.84653	0.85011	0.85348

Full T2-C improves upon T2-C μ -only for all three paired seeds in both overlaps. The corresponding reductions in mean joint error are 45.28% in $S-H$ and 26.16% in $H-P$. In the $S-H$ overlap, the μ -only gate remains close to the E2-probability blend, whereas in $H-P$ it provides a moderate correction. In both cases, incorporating the history descriptor yields a further and consistent improvement. These results support the use of history-conditioned assembly in the present centered-square setting.

Figures 3 and 4 provide complete views of the velocity, gauge-corrected pressure, and pointwise error. In the $S-H$ overlap, the Steady and Hopf predictions differ mainly in the near-wake recirculation region and in the downstream pressure-recovery pattern. The T2-C reconstruction remains visually close to the more accurate Hopf prediction while reducing localized discrepancies in both velocity and pressure, consistent with the error reductions reported in Tables 8 and 9.

In the $H-P$ overlap, the discrepancy between the Hopf and Periodic specialists is more pronounced around the obstacle corners, the separated shear layers, and the developing wake. The T2-C reconstruction reduces these localized errors while preserving the dominant large-scale wake structure, consistent with the substantially lower joint velocity–pressure errors reported for this overlap.

Full-window reevaluation and scalar controls. Table 12 uses all $K = 24$ steps, first averaging windows within each Reynolds number and then weighting the two held-out Reynolds numbers equally. It also reports the dimensionless squared relative objective, $L_{\text{sq}} = \text{mean}[(E_u/100)^2 + (E_p/100)^2]$. This is different from the displayed percentage joint error. The constant- α control minimizes the training-window squared objective analytically over $[0, 1]$; its weights are 0.357850555 in $S-H$ and 0.928683495 in $H-P$. The constant-logit control fits a single training-only offset in $[-20, 20]$, giving 0.490963974 and 3.480802826, respectively. The validation-fixed candidate is selected on validation squared error, rather than chosen a posteriori on held-out data. Both candidates are finite and no held-out windows are excluded in this reevaluation. These are deterministic scalar controls for the archived candidates.

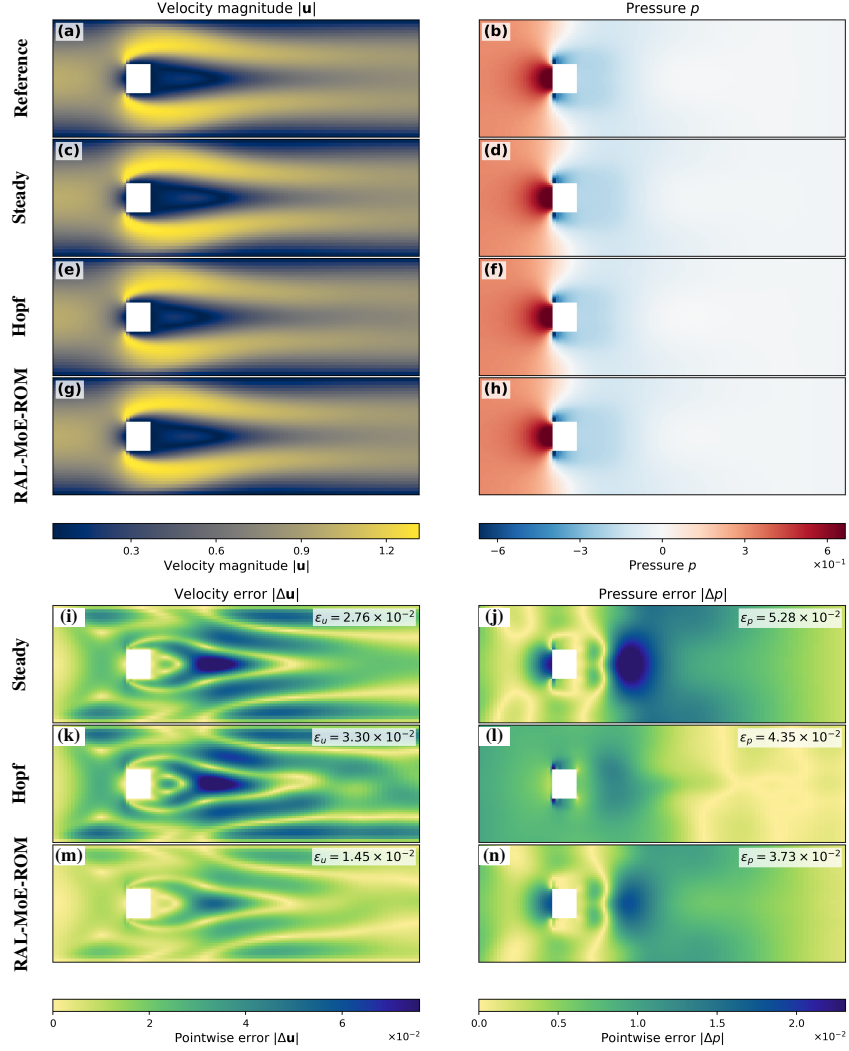


Figure 3: Representative held-out prediction in the centered-square $S-H$ overlap at $Re = 95.1$. The upper panel compares the archived reference reconstruction, Steady specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors. Panel identifiers run (a)–(h) for fields and (i)–(n) for errors. The displayed snapshots and color scales are retained from the archived figures; row labels and identifiers are typeset separately.

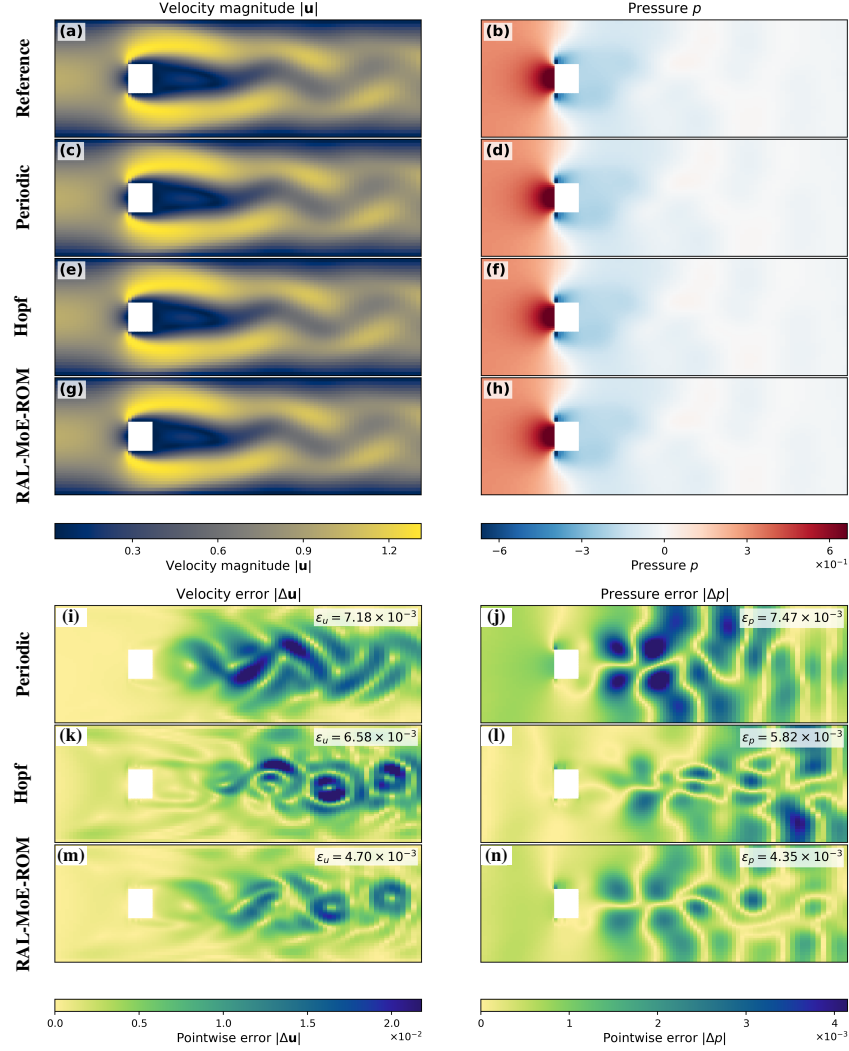


Figure 4: Representative held-out prediction in the centered-square H - P overlap at $Re = 100.5$. The upper panel compares the archived reference reconstruction, Periodic specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors. Panel identifiers run (a)–(h) for fields and (i)–(n) for errors. The displayed snapshots and color scales are retained from the archived figures; row labels and identifiers are typeset separately.

Table 12: Centered-square full-window $K = 24$ reevaluation using the archived physical-field quadratic cache. Errors are percentages against the cached POD reference; the last column is $10^4 L_{sq}$. This table uses a single frozen gate checkpoint, not the three-seed mean.

Overlap	Method	E_u	E_p	$E_u + E_p$	$10^4 L_{sq}$
$S-H$	E2 Top-1	0.5355	1.0432	1.5786	5.6364
	E2 probability	0.6624	1.1521	1.8145	4.3056
	Equal blend	1.0133	1.3717	2.3849	4.3665
	Train-fixed α	0.8128	1.2462	2.0591	4.1324
	Train-fixed logit offset	0.7857	1.2292	2.0149	4.1398
	Validation-fixed candidate	0.5355	1.0432	1.5786	5.6364
	T2-C	0.3831	0.7123	1.0954	2.3199
$H-P$	E2 Top-1	0.7263	0.9161	1.6424	4.8921
	E2 probability	0.6108	0.8381	1.4489	2.9087
	Equal blend	0.5920	0.8010	1.3930	2.5103
	Train-fixed α	0.5016	0.4752	0.9767	0.5818
	Train-fixed logit offset	0.5134	0.4687	0.9821	0.5941
	Validation-fixed candidate	0.5310	0.4758	1.0068	0.6323
	T2-C	0.4263	0.3860	0.8123	0.4072

Table 13: Three paired gate seeds: full-window joint error (%) at $K = 24$, mean $\pm$ sample standard deviation. All specialists and candidate trajectories are fixed; these are not full-pipeline seeds.

Overlap	μ -only	Full T2-C	Relative reduction
$S-H$	1.814699 ± 0.000242	1.095351 ± 0.000015	39.64%
$H-P$	0.976939 ± 0.000179	0.812343 ± 0.000177	16.85%

The single-checkpoint T2-C also improves on the training-fitted constant weight and constant-logit controls in both overlaps. Together with the paired μ -only ablation, this supports history-conditioned assembly for these fixed candidate sets; it does not establish long-time phase alignment or nonlinear physical consistency. Here “history-conditioned” means that the initial history can change the selected weight between windows; the weight remains fixed throughout each autonomous prediction window. A constant-logit offset can still vary its resulting weight with the E2 parameter preference, but cannot adapt to different histories at the same parameter. The controls therefore test more than a universal calibration bias, without proving that history dependence is necessary for every possible gate or flow regime.

F.2 ADDITIONAL PINBALL OVERLAP RESULTS

The formal outer candidates are Deep-FNN-H3 local predictors, not sparse-MoE specialists. These results concern physical-space assembly.

Table 14: Fluidic-pinball overlap joint errors (%). For $S-H$, Candidate 1 is the Steady specialist and Candidate 2 is Hopf; for $H-P$, Candidate 1 is the Periodic specialist and Candidate 2 is Hopf. The convex oracle selects its weight a posteriori for each evaluation window and is unavailable during prediction.

Overlap	K	Cand. 1	Cand. 2	E2-probability blend	T2-C	Convex oracle
$S-H$	24	0.6014	3.1925	1.7511	0.5897	0.5889
$S-H$	56	0.6927	3.1193	1.7714	0.6767	0.6763
$H-P$	24	0.0990	0.5146	0.2972	0.0945	0.0928
$H-P$	56	0.1028	0.5346	0.3085	0.0978	0.0955

T2-C improves upon Candidate 1 in all four overlap evaluations. The relative reductions in joint error are approximately 1.9% and 2.3% for $S-H$ at $K = 24$ and $K = 56$, and 4.5% and 4.9% for $H-P$,

respectively. Although these gains are more modest than those observed for the centered-square benchmark, they persist at both rollout horizons. T2-C also remains close to the corresponding window-wise convex oracle, providing a numerical comparison with the diagnostic convex reference, not a proof of optimality under other error aggregations.

G INTERNAL ROUTING AND PHYSICAL-FIELD CONSISTENCY DIAGNOSTICS

G.1 ADDITIONAL INTERNAL-ROUTING DIAGNOSTICS

Evaluation protocol and routing statistics. We analyze the frozen circular-cylinder Hopf and Periodic specialists on their held-out autonomous rollout windows. For the Periodic specialist, we use 13 fixed windows at each of four held-out Reynolds numbers, giving 52 windows of length $K = 48$. For the Hopf specialist, we use five fixed windows at each of three held-out Reynolds numbers, giving 15 windows of length $K = 56$. One routing record is collected per reduced time step at the first integration stage, resulting in 2496 Periodic and 840 Hopf routing decisions per channel. Thus the reported statistics characterize reduced time step routing rather than all intermediate right-hand-side evaluations. Overlapping windows are not interpreted as independent trajectories, and these diagnostics are not used for model selection.

For N routing decisions, let $I_{n,e} \in \{0, 1\}$ indicate whether routed expert e is selected at decision n , with the group identity included in the expert label. We define the normalized activation frequency

$$f_e = \frac{1}{2N} \sum_{n=1}^N I_{n,e}, \quad (53)$$

where the factor 2 accounts for the Top-2 routed experts selected at each decision, so that $\sum_e f_e = 1$. The corresponding mean routing mass is

$$m_e = \frac{1}{N} \sum_{n=1}^N \omega_{n,e} I_{n,e}, \quad (54)$$

where $\omega_{n,e}$ denotes the final routed mixing coefficient assigned to expert e . Experts that are never selected are retained with $f_e = m_e = 0$.

We summarize the dispersion of expert selection by the normalized utilization entropy

$$H = - \frac{\sum_{e: f_e > 0} f_e \log f_e}{\log N_E}, \quad 0 \leq H \leq 1, \quad (55)$$

where N_E is the number of routed experts available to the corresponding channel. Because H is computed from activation frequency rather than routing mass, it characterizes diversity of expert selection rather than equality of the assigned weights. Distinct Top-2 pairs retain group identity but are treated as unordered.

For the two specialists considered here, the architecture assigns a fixed coefficient $4/7$ to the shared expert and a total coefficient $3/7$ to the routed Top-2 component. Consequently,

$$\sum_e m_e = \frac{3}{7}. \quad (56)$$

The routing mass therefore measures mixture allocation; it does not quantify the norm of an expert output or, by itself, establish a physical interpretation. Labels such as $g_0 : e_0$ refer only to routed experts and should not be confused with the shared expert $e = 0$ in Eq. (12).

Table 15: Post-hoc internal-routing statistics for the circular-cylinder Hopf and Periodic specialists on held-out autonomous rollout windows. “Dominant-pair share” denotes the fraction of routing decisions associated with the most frequently selected unordered Top-2 pair. H denotes the normalized utilization entropy defined in Eq. (55).

Specialist	Channel	Active experts	Distinct Top-2 pairs	Dominant-pair share (%)	H
Hopf	Velocity	2/6	1	100.00	0.387
	Pressure	2/6	1	100.00	0.387
Periodic	Velocity	18/18	33	16.27	0.929
	Pressure	14/18	16	31.25	0.784

Periodic routing profile. The Periodic specialist uses a broad range of its available routed experts. Across the held-out rollouts, the velocity channel selects all 18 routed experts and realizes 33 distinct group–pair combinations, while the pressure channel selects 14 of 18 experts through 16 combinations (Table 15). The most frequent pair accounts for only 16.27% of velocity decisions and 31.25% of pressure decisions, consistent with the higher utilization entropy of the velocity channel. The selected groups also vary across the four held-out Reynolds numbers. Figure 5 summarizes the corresponding group frequencies and routed-expert masses.

Hopf routing profile. The Hopf specialist exhibits a substantially more concentrated routing pattern. Because it contains a single expert group, only channel-level routing is active. Across all evaluated windows, the velocity channel selects the pair (e_1, e_3) and the pressure channel selects (e_2, e_4) at every recorded reduced time step. A separate stage-wise check confirms that these same pairs are retained across all four Runge–Kutta stages.

Although the selected support is fixed, the routed weights are strongly nonuniform. Velocity expert e_3 receives approximately 99.93% of the total routed mass, while pressure expert e_2 receives 84.16%. The secondary pressure expert e_4 nevertheless receives a Reynolds-number-dependent weight, with its mean mass increasing across the three held-out parameter values.

The two specialists therefore exhibit different internal utilization patterns. The Periodic specialist distributes routing decisions across many group–pair combinations, whereas the Hopf specialist retains fixed sparse support and adapts primarily through the mixture weights. The single Hopf group is part of the prescribed architecture and should not be interpreted as collapse of a learned multi-group router.

Periodic rollout-position diagnostic. Figure 6 shows the mean routed-expert mass as a function of normalized rollout position, using the same 52 held-out Periodic windows. The horizontal coordinate reflects prediction position within the autonomous window rather than physical oscillation phase. This diagnostic complements the aggregate statistics above by showing how expert allocation varies over the autonomous rollout.

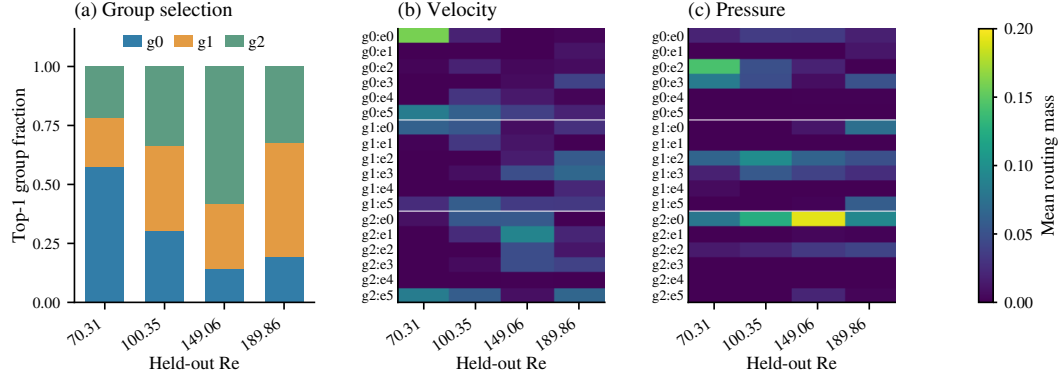


Figure 5: Internal routing of the circular-cylinder Periodic specialist on held-out autonomous roll-outs, recorded once per reduced time step at the first integration stage. (a) Top-1 group-selection frequencies. (b,c) Mean routed-expert mass for the velocity and pressure channels. Expert identifiers are local to the corresponding specialist, group, and channel and are not assigned physical labels.

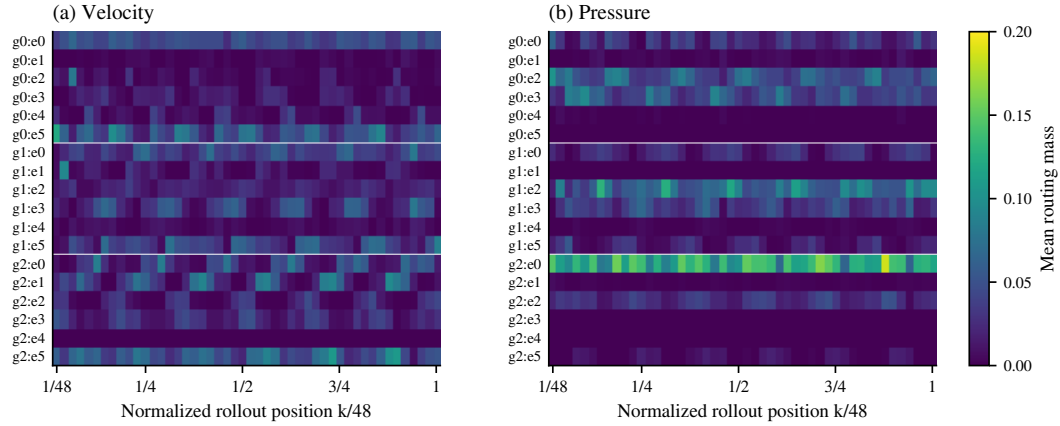


Figure 6: Routed-expert mass versus autonomous rollout position for the circular-cylinder Periodic specialist. Each column averages the routing decision at the corresponding reduced time step over the 52 fixed held-out windows, and all routed experts are retained. The horizontal coordinate is the normalized prediction position $k/48$ and is not interpreted as physical oscillation phase.

G.2 CONDITIONAL CONSISTENCY OF OUTPUT-SPACE FUSION

Let $\mathbf{u}_\alpha = \alpha \mathbf{u}_1 + \beta \mathbf{u}_2$ and $p_\alpha = \alpha p_1 + \beta p_2$, where $\beta = 1 - \alpha$, $\alpha \in [0, 1]$ is constant in space and time over the window, and both candidates share a mesh, parameter, time grid, and pressure gauge. For a common linear operator D_h , $D_h \mathbf{u}_\alpha = \alpha D_h \mathbf{u}_1 + \beta D_h \mathbf{u}_2$. Thus zero discrete divergence, a zero-mean pressure gauge, and identical affine boundary constraints are preserved only to the extent that both candidates satisfy them.

Nonlinear residuals. For a fixed bilinear discretization N_h , set $\delta \mathbf{u} = \mathbf{u}_1 - \mathbf{u}_2$. Bilinearity gives

$$N_h(\mathbf{u}_\alpha, \mathbf{u}_\alpha) = \alpha N_h(\mathbf{u}_1, \mathbf{u}_1) + \beta N_h(\mathbf{u}_2, \mathbf{u}_2) - \alpha \beta N_h(\delta \mathbf{u}, \delta \mathbf{u}). \quad (57)$$

With the sign convention $R_u(\mathbf{u}, p) = \partial_t \mathbf{u} - \mathbf{f}_h - L_h \mathbf{u} + N_h(\mathbf{u}, \mathbf{u}) + G_h p$, one obtains

$$R_u(\mathbf{u}_\alpha, p_\alpha) = \alpha R_u(\mathbf{u}_1, p_1) + \beta R_u(\mathbf{u}_2, p_2) - \alpha \beta N_h(\delta \mathbf{u}, \delta \mathbf{u}). \quad (58)$$

For the pressure residual $R_p(\mathbf{u}, p) = A_h p - [s_0 + S_1 \mathbf{u} + B_p(\mathbf{u}, \mathbf{u})]$, with fixed bilinear B_p , similarly

$$R_p(\mathbf{u}_\alpha, p_\alpha) = \alpha R_p(\mathbf{u}_1, p_1) + \beta R_p(\mathbf{u}_2, p_2) + \alpha \beta B_p(\delta \mathbf{u}, \delta \mathbf{u}). \quad (59)$$

These identities distinguish the candidates' existing residuals from the additional fusion defect. They are conditional algebraic identities, not measurements of the CFD residual: state-dependent limiters or other non-bilinear discretizations require their actual discrete operators.

Energy and phase cancellation. For $E(\mathbf{u}) = \frac{1}{2}\|\mathbf{u}\|_M^2$ with a common positive quadrature matrix,

$$E(\mathbf{u}_\alpha) = \alpha E(\mathbf{u}_1) + \beta E(\mathbf{u}_2) - \frac{\alpha\beta}{2}\|\mathbf{u}_1 - \mathbf{u}_2\|_M^2. \quad (60)$$

Candidate disagreement can therefore reduce the blend's energy relative to the weighted candidate energies, including through phase cancellation. This does not imply that its energy is below the reference solution or below both candidates. Since the blend is not fed back into either specialist, this term is not a recurrent dissipative modification of their reduced dynamics. Empirical residual and phase-sensitive evaluations remain separate from these identities.

G.2.1 MEASURED COMMON-GRID DIAGNOSTICS

We evaluate both candidates, their frozen T2-C blend, and the common POD-reconstructed reference on all 16 held-out windows at each square-cylinder interface ($K = 24$, output spacing 4). No window is selected by prediction quality. These are reconstructed-field diagnostics, not residuals of the original OpenFOAM finite-volume equations and not errors against unprojected CFD fields.

Operators and averaging. On the common 9,400-cell grid, quadratic local least-squares fits define D_x, D_y and L . The initial stencil has 24 nearest cells, with independent coordinate scaling; it is expanded only if the polynomial matrix is rank-deficient or has condition number above 10^6 . A 36-neighbor version checks sensitivity. Actual stencils contain at most 72 cells. Both versions differentiate a manufactured quadratic polynomial with maximum absolute error below 1.4×10^{-10} . We use

$$\hat{R}_u = \delta_t \mathbf{u} + N_D(\mathbf{u}, \mathbf{u}) + \nabla_D p - Re^{-1} L \mathbf{u}, \quad \hat{R}_p = Lp + D_x N_{D,x} + D_y N_{D,y}, \quad (61)$$

where $N_D(\mathbf{u}, \mathbf{u}) = u D_x \mathbf{u} + v D_y \mathbf{u}$. Residual norms use area-weighted RMS over the interior wake $6 \leq x \leq 16, 1 \leq y \leq 3$ (2,627 cells). Centered time differences and summary statistics use output steps 2–23. The nonzero reference residual includes POD truncation and spatial/temporal differentiation error; it is not a solver convergence tolerance. Windows are averaged within each parameter, then parameters equally. The two parameters per interface each contribute eight overlapping windows; these windows are not independent experimental replications.

Table 16: Measured square-interface diagnostics with the 24-neighbor initial stencil. Energy error is $|E - E_{\text{ref}}|/E_{\text{ref}}$ in percent over the whole domain; residuals are interior RMS. All columns average steps 2–23. Reference denotes POD-reconstructed CFD, not raw CFD.

Interface	Field	Energy error (%)	$\ \hat{R}_u\ $	$\ \hat{R}_p\ $
S–H	Reference	0	0.00460	0.03395
	Steady	2.81100	0.00530	0.03674
	Hopf	0.14296	0.00607	0.03513
	T2-C	0.17641	0.00520	0.03421
H–P	Reference	0	0.11486	0.03736
	Periodic	0.09163	0.11627	0.03785
	Hopf	0.17298	0.11189	0.04047
	T2-C	0.05974	0.11408	0.03631

Identities versus measured accuracy. The maximum absolute discrepancies in the energy, momentum-residual and pressure-residual identities are 1.2×10^{-15} , 7.4×10^{-15} and 6.1×10^{-14} , respectively, across both stencils and all windows. These values verify the implementation of the identities, not physical accuracy. The mean blend-energy deficit relative to reference energy is 0.00314% (S–H) and 0.00181% (H–P), distinct from the energy-error column in Table 16. T2-C has a lower measured pressure residual than either candidate for both interfaces and both stencils. However, it does not minimize every diagnostic: the Hopf candidate has a smaller H–P momentum

residual and a smaller S–H energy error. Changing the stencil also changes residual magnitudes substantially. Thus output blending is not a general residual minimizer, nor do these results establish discrete conservation or stability.

Phase diagnostics. A transverse-velocity probe is fixed geometrically at the cell nearest (8, 2.5). We detrend each signal and compare Hilbert phases with the reference. A window is marked unresolved if the reference dominant non-Nyquist Fourier component spans fewer than two cycles or its detrended standard deviation is below 10^{-5} . None of the 16 S–H windows and only 15 of the 16 H–P windows pass this screening. Even passing this necessary screen is not evidence that temporal aliasing or Hilbert endpoint effects are absent. We therefore report phase traces as short-window diagnostics, not as evidence for a general reduction of phase error or long-time phase stability.

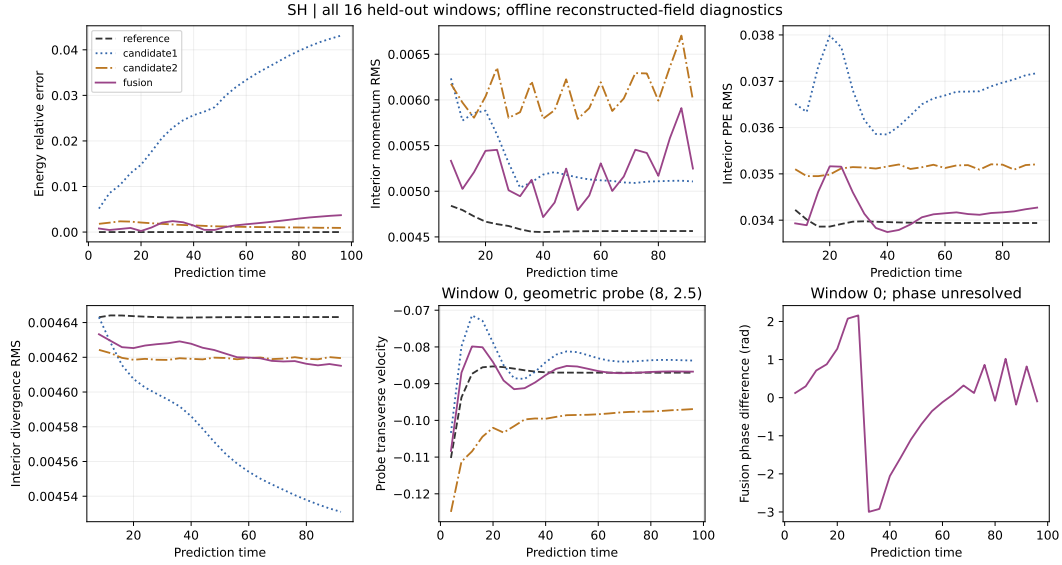


Figure 7: S–H diagnostics: the first four panels average all 16 windows; the last two show the first frozen window, not a best-case selection. Candidate 1 is Steady and candidate 2 is Hopf. Reference is POD-reconstructed CFD. The unresolved phase trace is descriptive only.

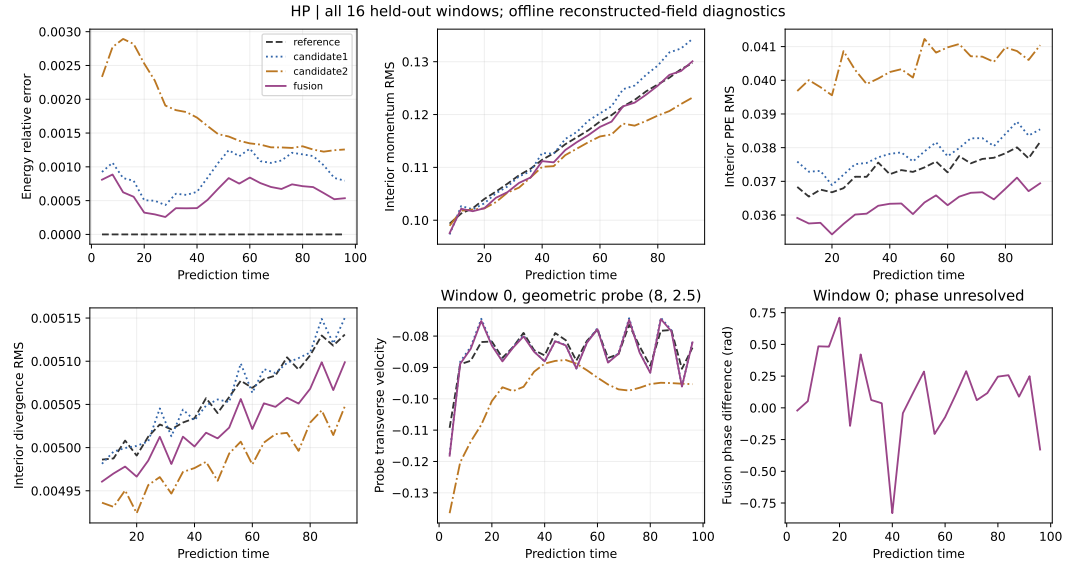


Figure 8: H-P diagnostics under the same protocol. Candidate 1 is Periodic and candidate 2 is Hopf. The first four panels average all 16 windows; the probe and phase panels show the first frozen window, which is unresolved by the stated phase screen.

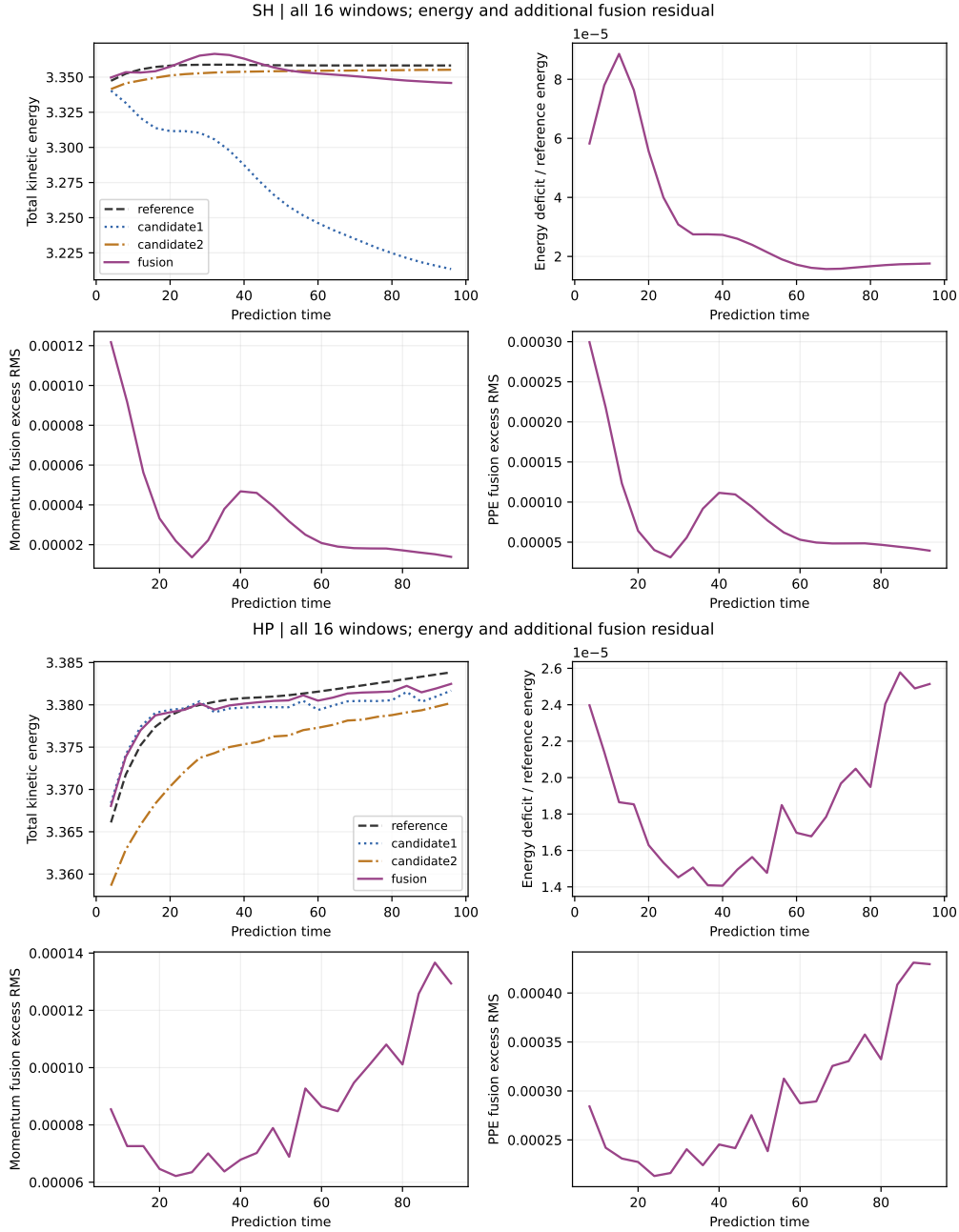


Figure 9: Actual kinetic energy and additional fusion defects averaged over all windows. The energy deficit is measured relative to the weighted candidate energies, normalized by reference energy; it is not the energy error. Momentum and pressure excesses are differences from the weighted candidate residuals, not the total fused residuals.

H COMPUTATIONAL COST AND TIMING BOUNDARIES

Counting convention. Total parameters count unique trainable parameters in each analyzed circular-cylinder specialist. Active parameters count unique trainable parameters used by one batch-size-one prediction-path routing decision, not their union across a batch, trajectory, or all RK4 stages. These include the encoder and refinement blocks, executed routers, the selected group’s shared experts, channel-specific Top-2 velocity and pressure experts, and the learned pressure cor-

rection. Fixed POD bases, projected operators, normalization statistics, and non-trainable buffers are excluded.

Table 17: Trainable and per-decision active parameter counts for the analyzed circular-cylinder specialists. Active counts refer to the sparse prediction path; counts are exact integers.

Specialist	Total trainable	Active prediction	Active fraction
Hopf	31,810,392	14,165,816	44.53%
Periodic	48,617,547	8,308,959	17.09%

The Hopf and Periodic specialists activate 44.53% and 17.09% of their trainable parameters per prediction decision, respectively. These counts describe conditional parameter activation, not measured FLOPs, runtime, or speedup. Complete parameter-matched counts are unavailable for the Vanilla-FNN-MoE, DataOnly-MoE, and Global MoE ablations, so the local architecture comparison is not interpreted as a capacity-matched study.

Native local latency diagnostic. For Circular Periodic seed 1248, batch size one and $K = 48$, we use three warm-up runs and ten timed repetitions. The native implementation includes modal roll-out, feature generation, data transfers, algebraic pressure, all-step physical decoding, and pressure gauge correction. It excludes physical-history encoding, E2, descriptor construction, T2-C fusion, model/data loading, and error computation. The Galerkin NumPy path executes on the CPU; sparse parameter activation does not imply a proportional wall-clock saving in this implementation.

Table 18: Local native Circular Periodic diagnostic. GPU memory is peak allocated bytes. These times are not complete end-to-end latencies.

Model	Median time (s)	GPU peak (bytes)	Time / Dense
Proposed	2.1848	204,238,336	3.32
Dense	0.6579	43,111,936	1.00
Structured	0.8652	43,316,736	1.32

Matching FOM logs alone does not supply an end-to-end speedup: the same geometry, physical prediction interval, initialization, and compute/I/O boundaries must also be measured for the complete ROM pipeline. No such speedup is claimed here.

Paired physical-input/physical-output specialist timing. A separate measurement adds full-size physical-history encoding and history right-hand-side construction to the original Periodic specialist (seed 1248). For Circular $Re = 70.314635$, the frozen first legal $K = 48$ window spans $t = 1384.49597$ to 1983.19690 , or 598.70093 physical time units. Its three-frame physical input is the POD-reconstructed history on 97,368 points; encoding is timed, but acquiring that history is not. All 48 velocity and pressure fields are decoded and pressure is gauge-corrected. The model and inputs are resident before timing.

On an RTX 4090 with a Xeon Platinum 8358P host, with one CPU numerical thread, batch size one, three warm-ups, ten repetitions, and CUDA synchronization, the median is 2.18747 s (mean 2.18729 s, sample standard deviation 0.00462 s; range 2.18034–2.19550 s). The matched production OpenFOAM log contains 4,692 solver steps and a ClockTime difference of 1,996 s; the largest endpoint discrepancy is 2.3×10^{-5} physical time units, attributable to stored timestamp precision. The production solver used one process on a Core i5-14600K VMware host with 16 vCPUs and at most five concurrent cases. Hardware identification is from the subsequent source audit, not a contemporaneous benchmark. There is one FOM timing realization, not ten repeated FOM runs.

The resulting ratio, $1996/2.18747 \simeq 912.5$, is specifically a historical production-FOM/warm-specialist wall-time ratio. The FOM interval includes native solver field writes but excludes meshing, initialization, periodic detection, and subsequent VTK/NPZ conversion. ROM model/input loading, output disk writes, E2 selection, and descriptor/T2-C operations are excluded. Different hardware,

1782 concurrency, and I/O boundaries prevent interpreting this ratio as a hardware-independent or com-
1783 plete hierarchical pipeline speedup. A fully paired Top-2 end-to-end timing remains unmeasured.
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